

# SIEVE LIMITS FOR THE ERDŐS–STRAUS EXCEPTIONAL SET: WHY $3/4$ IS SHARP FOR CONGRUENCE SIEVES

ANONYMOUS

ABSTRACT. Let  $E(N)$  count the  $n \leq N$  for which  $4/n$  is not a sum of three positive unit fractions. The best internally proved bound in this programme is  $E(N) \ll N \exp\{-c(\log N)^{3/4}\}$ . We show that the exponent  $3/4$  is a hard ceiling for a precisely defined class of congruence sieves. The class consists of nonnegative CRT majorants that are at least one on the whole avoider set of a family of forced congruence classes, whose final bound has the form  $N \cdot \mathbb{E}\nu + \sum_i |a_i|$ , and whose family primes are at most  $N^{O(1)}$ . The main theorem: for every fixed  $B$  and every family of the forced classes  $\mathcal{R}(M)$  with  $M \leq P(M)^{1+B}$  (twin moduli, prime-power tops and any number of prime factors at one scale included), every such sieve with the specified admissible set saves at most  $O_B((\log N)^{3/4})$ . The same holds for families in which every modulus has a dominant prime factor, including the  $(a, D)$ -classes and, modulo a published divisor-sum estimate of Elsholtz and Tao, the Case-A classes. No power of  $\log \log N$  can be gained. The key new tool is a weighted  $k$ -ary comparison theorem for sequential conditioned laws, proved by Lagrange extrapolation in artificial replacement coins and a supermartingale bound. The majorant of the  $3/4$  note lies in the class, and the large-sieve argument of the  $2/3$ -loglog note is capped in the same way at its own exponent. For finite prime-slice families the cap extends to per-frequency signed rounding with weights  $\geq 1$ , to majorants positive only at primes, and to CRT-evaluated moment methods. For Selberg-type majorants we prove a sieve limit in terms of noise stability, and a  $\Lambda^2$  cap of order  $(\log X)^{3/4+o(1)}$  at level  $O(\log X)$  for  $\mathcal{R}(M)$ -families with a fixed number of prime factors above  $(\log X)^8$ , with no condition relating  $M$  to  $P(M)$ ; its arithmetic input is a Brun–Titchmarsh bound for rough partners from the large sieve. We list exactly what the theorems exclude. Every statement carries its status label. All proofs were checked internally only; nothing here has been externally refereed. This is not a proof of the Erdős–Straus conjecture.

## 1. INTRODUCTION AND STATUS

Call  $n \geq 1$  *representable* if  $4/n = 1/x_1 + 1/x_2 + 1/x_3$  with positive integers  $x_i$ , and let  $E(N)$  count the non-representable  $n \leq N$ . The Erdős–Straus conjecture asserts that every  $n \geq 2$  is representable. Vaughan [V] proved  $E(N) \ll N \exp\{-c(\log N)^{2/3}\}$  with the large sieve (see the reconstruction of his argument in [PW, §4]; for the inequality see [M]). Two companion notes of this programme improve this. The first [LL] gains a factor  $(\log \log N)^{1/3}$  in the exponent. The second [TQ] proves

$$E(N) \ll N \exp\{-c(\log N)^{3/4}\},$$

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*Date:* Research note; internally reviewed results only, not externally refereed.

with label *internally proved; not externally refereed*. The 3/4 note itself explains 3/4 heuristically: the forced classes have cubic logarithmic mass, and a mass-driven optimisation gives  $\theta = b/(b+1)$  with mass exponent  $b = 3$  [TQ, §10].

This note turns that heuristic into a theorem for a precise class of sieves, and records exactly where the theorem stops.

**Status labels.** Every statement carries one of the following labels.

- [PROVED]: proved here, with the proof given or precisely cited to an internal document; checked by internal hostile review only.
- [PROVED MOD X]: proved assuming the published external result X, which was checked against the source but not re-proved.
- [CONDITIONAL]: proved assuming a named open hypothesis.
- [EVIDENCE]: numerical or model evidence; never a proof.
- *Assessment*: a reasoned judgement about methods or heuristics; not a theorem.
- [CONJECTURE] / [OPEN]: not proved.

Apart from the classical facts of Section 2, each [PROVED] result below appears in one of eight internal working documents, [ET], [EB], [TW], [TW2], [TW3], [KA], [TW4], [NC]. Each was hostile-reviewed; the reviews are [ETr, EBr, TWr, TW2r, TW3r, KAr, KAr2, TW4r, NCr]. No prior source for the sieve-limit theorem (Theorem 4.5) was found; it may be folklore in spirit. Novelty is not claimed.

## Results.

- (1) *The main theorem* (Theorems 9.3 and 9.4, [PROVED]). Fix  $B$ . For every family of classes  $\mathcal{R}(M)$  with  $M \leq P(M)^{1+B}$ , of any shape (dominant, gapped, twin, prime-power top, any number of prime factors at comparable scale), every nonnegative majorant of level  $\lambda$  of the avoider set saves at most  $C(B)\lambda^{3/4}$ . Hence every CRT-majorant architecture (Definition 3.3:  $\nu \geq 0$  on  $\mathbb{Z}$ ,  $\nu \geq 1$  on the whole avoider set, final bound  $N\mathbb{E}\nu + \sum |a_i|$ , family primes  $\leq N^{O(1)}$ ) for such a family, with the admissible set specified there, saves at most  $O_B((\log N)^{3/4})$ . The same holds for families whose moduli all have a dominant prime, Case-A and  $(a, D)$ -classes included (Case A [PROVED MOD] Elsholtz–Tao Prop. 1.4).
- (2) *The tools*. The sieve-limit theorem for prime-slice systems (Theorem 4.5): every majorant of level  $\lambda$  saves at most  $C \inf_{\alpha} [\alpha\lambda + \sum_{\ell} \bar{p}_{\ell}\ell^{-\alpha}]$  beyond an admissible-set term, plus an error term that is  $O(\log^2 \lambda)$  in all applications here (Remark 4.6); its sequential version (Theorem 4.9); the cubic profile  $\sum \bar{p}_{\ell}\ell^{-\alpha} \ll \alpha^{-3}$  of the forced classes (Section 5); the quadratic-residue base built from Mordell’s obstruction and the capped sequential measure (Section 7); and a weighted  $k$ -ary comparison theorem (Theorem 8.5), which replaces the one-coordinate-per-window hypothesis of the sequential sieve limit. It couples the sequential conditioned law with a product draw, freezes the path, extrapolates in artificial Bernoulli replacement coins by Lagrange interpolation for binomial laws (Lemma 8.3), and pays for the thinning with a supermartingale (Lemma 8.4). Only the mean activated mass enters, because the sequential sieve limit accepts random step costs (Theorem 9.1).
- (3) *The campaign sieves are in the class* (Section 10, [PROVED]). The majorant of the 3/4 note is in the class, so 3/4 is sharp for its architecture; the

large sieve of the  $2/3$ -loglog note is capped at its own exponent  $\frac{2}{3}$  with the  $(\log \log N)^{1/3}$  factor.

- (4) *Selberg-type majorants* (Section 11). A  $\Lambda^2$  sieve limit for arbitrary systems in terms of noise stability (Theorem 11.1, [PROVED]), and  $\Lambda^2$  caps  $\ll L^{3/4}(\log L)^{O_r(1)}$  for  $\mathcal{R}(M)$ -families with at most  $r$  prime factors above  $(\log X)^8$ ,  $r$  fixed, with (Theorem 11.11) or without (Theorem 11.12) the condition  $M \leq P(M)^{1+B}$  ([PROVED]). Their new arithmetic input is a Brun–Titchmarsh bound for rough partners from the arithmetic large sieve (Lemma 11.10). Under the  $B$ -hypothesis these caps are superseded by the main theorem (Remark 11.8).
- (5) *Beyond CRT evaluation* (Section 12, [PROVED]). For the finite prime-slice families, the cap extends to every rounding bound  $\sum_{\theta \neq 0} |\hat{\nu}(\theta)|w(\theta)$  with weight  $w \geq 1$ , with no bound on the size of the slice primes (Theorem 12.3); to majorants positive only at primes, hence to prime equidistribution at any level  $N^{O(1)}$  (Theorem 12.4); and to moment methods whose moments are evaluated through CRT densities.
- (6) *Exclusions and open problems* (Sections 13 and 14). A precise list of what lies outside: chiefly  $\mathcal{R}(M)$ -moduli with  $\log M / \log P(M)$  unbounded for general majorants,  $(a, D)$ - and Case-A moduli without a dominant prime, non-selector admissible sets, rounding with weights below one and cancellation between frequencies, and  $H_{\text{MS}}$ .

None of this bounds  $E(N)$  from below or says anything about methods outside the class; in particular it is not evidence against the conjecture.

## 2. FORCED CLASSES

A residue class  $b \pmod{G}$  is *forced* if every positive integer  $n \equiv b \pmod{G}$  is representable. We use three families of forced classes. For an integer  $D \geq 1$  put  $g(D) = \prod_p p^{\lceil v_p(D)/2 \rceil}$ , so that  $g(D) \mid D \mid g(D)^2$  and, for every integer  $x$ ,  $D \mid x^2 \iff g(D) \mid x$ .

**Lemma 2.1** (multiplier identity; [LL, Lemma 2.1]; [PROVED]). *Let  $M \equiv 3 \pmod{4}$ ,  $A = (M+1)/4$ , and  $A = uvw$  with positive integers  $u, v, w$ . If  $n \geq 1$  and  $nv \equiv -u \pmod{M}$ , then  $s = (nv+u)/M$  is a positive integer and  $4/n = 1/(suw) + 1/(nsvw) + 1/(nuvw)$ .*

*Proof.* On the common denominator  $nsuvw$  the numerators sum to  $nv + u + s = s(M+1) = 4suvw$ .  $\square$

**Lemma 2.2** (the classes  $\mathcal{R}(M)$ ; [N, Lemma 18.1]; [PROVED]). *For  $M \equiv 3 \pmod{4}$  let  $\mathcal{R}(M)$  be the set of classes  $-uv^{-1} \pmod{M}$  over all factorisations  $A = uvw$ . Then*

$$\mathcal{R}(M) = \{-4D \pmod{M} : D \mid A^2\}, \quad |\mathcal{R}(M)| \leq \tau(A^2),$$

*and every class of  $\mathcal{R}(M)$  is forced.*

*Proof.* Since  $4A \equiv 1 \pmod{M}$ , a factorisation  $A = uvw$  gives  $-uv^{-1} \equiv -u^2wA^{-1} \equiv -4u^2w \pmod{M}$  with  $u^2w \mid A^2$ . Conversely, if  $A = \prod p^{e_p}$  and  $D = \prod p^{d_p}$  with  $d_p \leq 2e_p$ , put  $v_p(u) = \lfloor d_p/2 \rfloor$ ,  $v_p(w) = d_p - 2\lfloor d_p/2 \rfloor$  and  $v_p(v) = e_p - v_p(u) - v_p(w) \geq 0$ ; then  $uvw = A$  and  $u^2w = D$ . Forcedness is Lemma 2.1.  $\square$

**Lemma 2.3** ( $(a, D)$ -classes; [ET, Lemma 3.2]; [PROVED]). *For all integers  $a, D \geq 1$  the class  $n \equiv -(4D + a) \pmod{4ag(D)}$  is forced.*

*Proof.* Write  $n + 4D + a = 4ag(D)j$  with  $j \geq 1$ . Then  $M := (n + 4D)/a = 4g(D)j - 1$  is  $\equiv 3 \pmod{4}$  and  $(M + 1)/4 = g(D)j$ , so  $D \mid g(D)^2 \mid ((M + 1)/4)^2$ . Since  $n \equiv -4D \pmod{M}$ ,  $n$  lies in a class of  $\mathcal{R}(M)$ , and Lemma 2.2 applies.  $\square$

These classes encode exactly the Case-B witnesses of the classical parametrisation [N, Thm 3.1(B)] (see [ET, Lemma 3.2]).

**Lemma 2.4** (Case-A classes; [ET, §3]; [PROVED]). *For  $d \geq 1$  and  $m \mid 4d + 1$  (so  $m$  is odd and  $(m, 4g(d)) = 1$ ), the class  $n \equiv -m^{-1} \pmod{4g(d)}$  is forced.*

*Proof.* Put  $z_0 = (nm + 1)/4$ , an integer. Then  $g(d) \mid z_0$ , so  $d \mid z_0^2$ ; and  $m \mid 4d + 1$  gives  $m \mid 4d + 4z_0$ , so  $m \mid d + z_0$ . With  $x = (z_0 + d)/m$  and  $y = (z_0 + z_0^2/d)/m = z_0(d + z_0)/(dm)$ , both integers since  $(m, d) = 1$ , one has  $1/x + 1/y = m/z_0$ , hence  $1/x + 1/y + 1/(nz_0) = (nm + 1)/(nz_0) = 4/n$ .  $\square$

We write *Case-B classes* for the classes of  $\mathcal{R}(M)$  and the  $(a, D)$ -classes, and *Case-A classes* for those of Lemma 2.4. Two facts are used repeatedly.

**Lemma 2.5** ( $n = 1$  is never covered; [PROVED]). *No forced class contains  $n = 1$ .*

*Proof.*  $4/1 > 3 \geq 1/x_1 + 1/x_2 + 1/x_3$ .  $\square$

**Lemma 2.6** (Mordell's obstruction; classical; [TW, Lemma 1.1]; [PROVED]). *Let  $M \equiv 3 \pmod{4}$ ,  $A = (M + 1)/4$  and  $D \mid A^2$ . Then  $(D, M) = 1$  and the Jacobi symbol satisfies  $(\frac{-4D}{M}) = -1$ . Consequently, if  $n$  is a nonzero quadratic residue modulo every prime factor of  $M$ , then  $n$  lies in no class of  $\mathcal{R}(M)$ .*

*Proof.*  $(A, M) = (A, 4A - 1) = 1$ . Write  $D = sr^2$  with  $s$  squarefree; then  $D \mid A^2$  gives  $sr \mid A$ , so  $M \equiv -1 \pmod{4s}$ . Now  $(\frac{-4D}{M}) = (\frac{-1}{M})(\frac{s}{M}) = -(\frac{s}{M})$ . If  $s$  is odd, reciprocity and  $(\frac{M}{s}) = (\frac{-1}{s})$  give  $(\frac{s}{M}) = 1$ , since  $(M - 1)/2$  is odd. If  $s = 2s'$ , then  $8 \mid M + 1$ , so  $(\frac{2}{M}) = 1$ , and  $(\frac{s'}{M}) = 1$  as before. For the consequence,  $(\frac{n}{M}) = \prod_{p \mid M} (\frac{n}{p}) = 1$ .  $\square$

### 3. CONGRUENCE SIEVES: THE ARCHITECTURE CLASS

#### 3.1. CRT systems and majorants.

**Definition 3.1** (prime-slice CRT system [ET, §1]). The data are a modulus  $Q_0$ ; a finite set  $\mathcal{P}$  of primes not dividing  $Q_0$  (the *slice primes*); a set  $R \subseteq \mathbb{Z}/Q_0$  of admissible small residues; and, for each  $c \in R$  and  $\ell \in \mathcal{P}$ , a set  $F_\ell(c) \subseteq \mathbb{Z}/\ell$  of forbidden classes. The *avoider set* is

$$\mathcal{A} = \{n : n \bmod Q_0 \in R, n \bmod \ell \notin F_\ell(n \bmod Q_0) \text{ for all } \ell \in \mathcal{P}\}.$$

Put  $p_\ell(c) = |F_\ell(c)|/\ell$  and  $\bar{p}_\ell = |R|^{-1} \sum_{c \in R} p_\ell(c)$ .

A family of conditions  $n \equiv b \pmod{q_0 \ell}$  with  $q_0 \mid Q_0$  and  $\ell \in \mathcal{P}$  induces such a system with  $F_\ell(c) = \{b \bmod \ell : b \equiv c \pmod{q_0}\}$ . A typical  $R$  is a *selector*  $R = \{c : (c, P) = 1\}$  with  $P \mid Q_0$ .

**Definition 3.2** (majorant of level  $\lambda$ ). A *majorant* of  $\mathcal{A}$  is a finite real combination

$$\nu(n) = \sum_i a_i \mathbf{1}[n \equiv b_i \pmod{d_i}]$$

with  $\nu \geq 0$  on all of  $\mathbb{Z}$  and  $\nu \geq 1$  on  $\mathcal{A}$ . It has *level*  $\lambda$  if  $\sum_{\ell \in \mathcal{P}, \ell | d_i} \log \ell \leq \lambda$  for every  $i$ . Its *mean* is  $\mathbb{E}\nu = \sum_i a_i/d_i$ , the average over a common period, and its *saving* is  $\log(1/\mathbb{E}\nu)$ .

The level charges only the slice primes; the small part of  $d_i$  is free. This is generous to the sieve. For every  $N$ ,

$$\sum_{n \leq N} \nu(n) = N \cdot \mathbb{E}\nu + O\left(\sum_i |a_i|\right), \quad (1)$$

since each class meets  $[1, N]$  in  $N/d_i + O(1)$  points.

### 3.2. The architecture class.

**Definition 3.3** (CRT-majorant architecture). A *CRT-majorant architecture* for a family  $\mathfrak{F}$  of forced classes bounds  $\#(\mathcal{A} \cap [1, N])$ , and hence the exceptional integers in  $[1, N]$  that satisfy the admissibility condition  $n \bmod Q_0 \in R$ , as follows.

- (A1)  $\nu$  is a majorant of the avoider set  $\mathcal{A}$  of  $\mathfrak{F}$  in the sense of Definition 3.2:  $\nu \geq 0$  on  $\mathbb{Z}$  and  $\nu \geq 1$  on the *whole* avoider set, not only on exceptional primes.
- (A2) The final bound is  $N \cdot \mathbb{E}\nu + T$  with  $T = \sum_i |a_i|$ , and it is non-trivial:  $T < N$ .
- (A3) The slice primes of the family are at most  $N^A$  for a fixed  $A$ .

Every sieve majorant in use satisfies (A1): Bonferroni truncations  $Q_r(H) = \binom{H-1}{r}$  and Selberg squares. Both campaign notes satisfy (A1)–(A3) (Section 10).

**Lemma 3.4** (coefficient budget implies level; [ET, Lemma 2.9], from [ETr, item 3]; [PROVED]). *Take a prime-slice system, or the sequential setting of Section 4.4. Let  $\nu$  be a majorant of  $\mathcal{A}$  with  $T = \sum_i |a_i|$ , and assume that every slice prime satisfies  $\log \ell \leq \Lambda_0$ . Then for every  $\lambda$  there is a majorant  $\nu'$  of level at most  $\lambda$  with  $\nu' \geq \nu$  pointwise and  $\mathbb{E}\nu' \leq \mathbb{E}\nu + T e^{\Lambda_0 - \lambda}$ . In particular  $\lambda = \Lambda_0 + \log T + \log(1/\mathbb{E}\nu)$  gives  $\mathbb{E}\nu' \leq 2\mathbb{E}\nu$ .*

*Proof.* Keep every term of level at most  $\lambda$ . Drop each term of level  $> \lambda$  with  $a_i < 0$ ; this raises  $\nu$  pointwise and raises the mean by  $|a_i|/d_i \leq |a_i|e^{-\lambda}$ . For a term of level  $> \lambda$  with  $a_i > 0$ , list its slice primes in increasing order and keep the longest prefix of level at most  $\lambda$ . The next prime has  $\log \ell \leq \Lambda_0$ , so the prefix has level  $> \lambda - \Lambda_0$ . Replace  $d_i$  by the divisor  $d'_i$  formed by the non-slice part of  $d_i$  and the kept prime powers. Then  $\mathbf{1}[n \equiv b_i \pmod{d'_i}] \geq \mathbf{1}[n \equiv b_i \pmod{d_i}]$ , and the mean rises by at most  $a_i/d'_i \leq a_i e^{\Lambda_0 - \lambda}$ . So  $\nu' \geq \nu$ ,  $\nu'$  is a majorant, and summing gives the bound.  $\square$

**Corollary 3.5** ([PROVED]). *Assume (A2)–(A3) and put  $s = \log(1/\mathbb{E}\nu)$ . Then there is a majorant  $\nu' \geq \nu$  of level at most  $(A+2)\log N$  with  $\mathbb{E}\nu' \leq \mathbb{E}\nu + N^{-1}$ . Hence, if every majorant of level at most  $(A+2)\log N$  has saving at most  $C(\log N)^{3/4} + C'$ , then  $s \leq \log 2 + C(\log N)^{3/4} + C'$  for all  $N \geq N_0(C, C')$ .*

*Proof.* Lemma 3.4 with  $\Lambda_0 = A \log N$ ,  $\lambda = (A+2)\log N$  and  $T < N$  gives  $\mathbb{E}\nu' \leq \mathbb{E}\nu + N \cdot N^A \cdot N^{-A-2} \leq \mathbb{E}\nu + N^{-1}$ . So  $\log(1/\mathbb{E}\nu') \geq \min(s, \log N) - \log 2$ , and the hypothesis gives  $\min(s, \log N) \leq \log 2 + C(\log N)^{3/4} + C'$ . For  $N \geq N_0$  the right side is  $< \log N$ , so the minimum is  $s$ .  $\square$

This form needs the cap only at level  $(A+2)\log N$ .

*Remark 3.6* (positivity on all of  $\mathbb{Z}$ ). Condition (A1) cannot be weakened to positivity on  $[1, N]$ . Restricted to  $[1, N]$ , the class  $\mathbf{1}[n \equiv b \pmod{N}]$  is the point mass at  $b$ , so the span of level-log  $N$  class indicators contains every function on  $[1, N]$ . The relaxed linear programme is then the exact count, and evaluating it presupposes knowing the avoider set [ET, §2.4]. Such methods are not sieves in the present sense.

#### 4. THE SIEVE-LIMIT THEOREM

All results of this section are from [ET, §2]; the review [ETr, items 1–2] found them sound and checked every intermediate inequality against exact linear programmes on small systems. Write  $C_4 = 8e^5$ .

##### 4.1. Two one-dimensional lemmas.

**Lemma 4.1** (binomial mass; [PROVED]). *Let  $\psi = \text{Bin}(z, q)$  with  $0 < q \leq 1/4$  and  $\mu' = zq$ . For every integer  $1 \leq y \leq z - 1$ ,  $\psi(y) \geq (8y)^{-1/2} \exp\{-\frac{4}{3}(y - \mu')^2/\mu'\}$ .*

*Proof.* By [MS, Ch. 10, Lemma 7],  $\binom{z}{y} \geq (8y(1 - y/z))^{-1/2} e^{zH(y/z)}$ , so  $\psi(y) \geq (8y)^{-1/2} e^{-zD(y/z||q)}$ . The bound  $D(a||q) \leq (a - q)^2/(q(1 - q))$  gives  $zD \leq (y - \mu')^2/(\mu'(1 - q)) \leq \frac{4}{3}(y - \mu')^2/\mu'$ .  $\square$

**Lemma 4.2** (node lemma; [PROVED]). *Keep  $\psi, q, \mu'$  as in Lemma 4.1. For a set  $Y$  of  $k + 1$  distinct integers in  $[0, z]$  let  $B(Y) = \max_{y \in Y} |\ell_y^Y(0)|/\psi(y)$ , where  $\ell_y^Y$  is the Lagrange basis polynomial of  $Y$ . Then every polynomial  $P$  of degree at most  $k$  with  $P \geq 0$  on  $Y$  satisfies  $|P(0)| \leq B(Y) \sum_{y \in Y} \psi(y)P(y)$ . Let  $B(k) = \min_Y B(Y)$  and  $C_1 = 16e^6$ . Then*

- (a)  $\log B(0) \leq \frac{1}{2} \log(8\mu' + 8) + 2$ ;
- (b)  $\log B(k) \leq 4 \log(4/3)\mu' \leq 1.151\mu'$  for every  $k \leq z$ ;
- (c) if  $\mu' \geq 64$  and  $1 \leq k \leq \mu'/16$ , then  $\log B(k) \leq \frac{k}{2} \log(C_1\mu'/k) + \frac{1}{2} \log(16\mu')$ .

Consequently, for all  $\alpha, s > 0$  and  $0 \leq k \leq z$ ,

$$\log B(k) \leq 19\alpha ks + C_4\mu'e^{-2\alpha s} + \frac{1}{2} \log(16\mu' + 16) + 74. \quad (2)$$

*Proof.* The inequality for  $P$  is Lagrange interpolation  $P(0) = \sum_y \ell_y(0)P(y)$  with  $P \geq 0$  on  $Y$ . (a) If  $\mu' \geq 1$  then  $z \geq 4$ ; take the single node  $y = \lceil \mu' \rceil \leq z - 1$  and apply Lemma 4.1 with  $|y - \mu'| \leq 1$ . If  $\mu' < 1$  take the node 0:  $\psi(0) = (1 - q)^z \geq e^{-1.151\mu'}$ . (b) Take  $Y = \{0, \dots, k\}$ ; then  $\ell_y(0) = \delta_{y0}$  and  $B = 1/\psi(0) \leq e^{1.151\mu'}$ . (c) Take nodes  $y_i = a + ih$ ,  $0 \leq i \leq k$ , with  $W = \lceil \sqrt{k\mu'}/2 \rceil$ ,  $h = \lceil 2W/k \rceil$ ,  $a = \lceil \mu' \rceil - W$ . Since  $k \leq \mu'/16$  and  $\mu' \geq 64$ , every node lies in  $[\mu'/2, 2\mu'] \subseteq [1, z - 1]$  and  $|y_i - \mu'| \leq \sqrt{k\mu'}$ , so Lemma 4.1 gives  $\psi(y_i) \geq (16\mu')^{-1/2} e^{-4k/3}$ . Next  $|\ell_i(0)| = \prod_{j \neq i} y_j / (h^k i! (k - i)!)$ , and  $\prod_{j \neq i} y_j \leq (2\mu')^k$ ,  $k!/(i!(k - i)!) \leq 2^k$ ,  $k! \geq (k/e)^k$  and  $h \geq \sqrt{\mu'/k}$  give  $|\ell_i(0)| \leq (4e\sqrt{\mu'/k})^k$ . Multiplying gives (c), since  $16e^{2+8/3} \leq C_1$ .

For (2): when (c) applies, use  $\frac{k}{2} \log(C_1\mu'/k) \leq \alpha ks + (C_1\mu'/2e)e^{-2\alpha s}$  (the maximum over real  $k$  of the difference), and  $C_1/(2e) = C_4$ . Otherwise use (a) or (b): if  $\mu' < 64$  then  $1.151\mu' \leq 74$ ; if  $k > \mu'/16$  and  $\alpha s \geq 1$  then  $1.151\mu' \leq 18.5k \leq 19\alpha ks$ ; if  $\alpha s < 1$  then  $1.151\mu' \leq 8.6\mu'e^{-2\alpha s}$ .  $\square$

**Lemma 4.3** (combination technique; standard; [PROVED]). *Let  $\Lambda \subset \mathbb{Z}_{\geq 0}^G$  be a lower set and  $P_\Lambda$  the span of the monomials  $x^a$ ,  $a \in \Lambda$ . For each coordinate  $g$  and degree  $k$  let  $I_k^{(g)}$  be Lagrange interpolation at any  $k + 1$  distinct nodes, and*

put  $c_j = \sum_{e \in \{0,1\}^G, j+e \in \Lambda} (-1)^{|e|}$ . Then for  $Q \in P_\Lambda$ ,  $Q = \sum_{j \in \Lambda} c_j (\bigotimes_g I_{j_g}^{(g)}) Q$ , and  $|c_j| \leq 2^G$ .

*Proof.* Put  $\Delta_k = I_k - I_{k-1}$  with  $I_{-1} = 0$ ; expanding  $\sum_{j \in \Lambda} \bigotimes_g \Delta_{j_g}$  gives the coefficients  $c_j$ . For  $a \in \Lambda$ ,  $\Delta_{j_g} x_g^{a_g} = 0$  when  $j_g > a_g$ , and the box  $\{j \leq a\}$  lies in  $\Lambda$ , so the sum applied to  $x^a$  is  $\prod_g \sum_{j_g \leq a_g} \Delta_{j_g} x_g^{a_g} = x^a$ .  $\square$

**4.2. The Boolean proposition.** Let  $x_i \sim \text{Bern}(p_i)$ ,  $i \in I$ , be independent, with weights  $s_i \geq s_* > 0$ . A function  $f$  on  $\{0,1\}^I$  is  $\lambda$ -level if  $f = \sum_T f_T$  where  $f_T$  depends only on  $x_T$  and  $\sum_{i \in T} s_i \leq \lambda$ .

**Proposition 4.4** ([ET, Prop. 2.4]; [PROVED]). Assume  $p_i \leq 1/4$  whenever  $s_i \leq \lambda$ , and  $p_i < 1$  for all  $i$ . Let  $f$  be  $\lambda$ -level with  $f \geq 0$  and  $f(0) \geq 1$ , and let  $\lambda \geq s_*$ . Put  $G = \lfloor \log_2(\lambda/s_*) \rfloor + 1$  and  $\mu = \sum_{s_i \leq \lambda} p_i$ . Then for every  $\alpha > 0$ ,

$$\log \frac{1}{\mathbb{E}f} \leq 19\alpha\lambda + C_4 \sum_{s_i \leq \lambda} p_i e^{-\alpha s_i} + G(75 + \log(2 + \lambda/s_*)) + \frac{G}{2} \log(16\mu + 16). \quad (3)$$

*Proof.* *Step 0.* No  $f_T$  contains an  $i$  with  $s_i > \lambda$ , so  $f$  ignores those coordinates; discard them, coordinates with  $p_i = 0$ , and empty bands.

*Step 1 (bands).* Put  $B_g = \{i : 2^g s_* \leq s_i < 2^{g+1} s_*\}$  for  $0 \leq g < G$ ,  $s_g = 2^g s_*$  and  $q_g = \max_{i \in B_g} p_i \leq 1/4$ .

*Step 2 (thinning).* Take independent  $u_i \sim \text{Bern}(q_g)$  and  $w_i \sim \text{Bern}(p_i/q_g)$ ; then  $u_i w_i$  has the law of  $x_i$ . Fix  $w$ , let  $Z_g = \{i \in B_g : w_i = 1\}$  and  $z_g = |Z_g|$ . Then  $f_w(u) = f(u \circ w)$  depends only on  $u_Z$ , is  $\lambda$ -level and nonnegative, and  $f_w(0) \geq 1$ .

*Step 3 (symmetrisation).* Average  $f_w$  over  $\prod_g \text{Sym}(Z_g)$ , which preserves the law of  $u$ . Each monomial  $u^S$  of a term  $f_T$  has multidegree  $(|S \cap Z_g|)_g$  in the lower set  $\Lambda = \{j : \sum_g j_g s_g \leq \lambda\}$ , and symmetrises to  $\prod_g \binom{K_g}{|S_g|} / \binom{z_g}{|S_g|}$  with  $K_g = |u_{Z_g}|$ . Hence  $\mathbb{E}_u f_w = \mathbb{E} Q(K)$  where  $Q \in P_\Lambda$ , the  $K_g \sim \text{Bin}(z_g, q_g)$  are independent,  $Q \geq 0$  on  $\prod_g \{0, \dots, z_g\}$  and  $Q(0) \geq 1$ . Reducing modulo  $\prod_{i=0}^{z_g} (K_g - i)$  we may take  $Q \in P_{\Lambda'}$ ,  $\Lambda' = \Lambda \cap \prod_g [0, z_g]$ .

*Step 4 (interpolation).* Choose the node sets of Lemma 4.2 inside  $\{0, \dots, z_g\}$  and apply Lemma 4.3 at 0. All grid points are support points, where  $Q \geq 0$ , so

$$1 \leq Q(0) \leq \left( 2^G |\Lambda'| \max_{j \in \Lambda'} \prod_g B_g(j_g) \right) \mathbb{E} Q(K).$$

(If  $z_g = 0$  then  $j_g = 0$  on  $\Lambda'$  and  $B_g(0) = 1$ .)

*Step 5 (bounds).* Apply (2) with  $s = s_g$  and  $\mu' = q_g z_g$ , using  $\sum_g j_g s_g \leq \lambda$  and  $|\Lambda| \leq (1 + \lambda/s_*)^G$ :  $\log(1/\mathbb{E}_u f_w) \leq \Phi(w)$ , where  $\Phi(w) = 19\alpha\lambda + \sum_g [C_4 q_g z_g e^{-2\alpha s_g} + \frac{1}{2} \log(16q_g z_g + 16) + 74] + G \log 2 + G \log(1 + \lambda/s_*)$ . By Jensen,  $\mathbb{E}f = \mathbb{E}_w \mathbb{E}_u f_w \geq e^{-\mathbb{E}_w \Phi(w)}$ . Finally  $\mathbb{E}_w q_g z_g = \mu_g := \sum_{i \in B_g} p_i$ , concavity of  $\log$ ,  $e^{-2\alpha s_g} \leq e^{-\alpha s_i}$  on  $B_g$ , and  $\sum_g \log(16\mu_g + 16) \leq G \log(16\mu + 16)$ .  $\square$

### 4.3. The CRT form.

**Theorem 4.5** (sieve limit for prime-slice systems; [ET, Thm 2.5]; [PROVED]). Take a prime-slice system. Assume that for every  $c \in R$ ,  $|F_\ell(c)| \leq \ell/4$  for every  $\ell \in \mathcal{P}$  with  $\ell \leq e^\lambda$ , and  $|F_\ell(c)| < \ell$  for every  $\ell \in \mathcal{P}$ . Let  $\bar{\mu} = \sum_{\ell \in \mathcal{P}, \ell \leq e^\lambda} \bar{p}_\ell$  be the truncated mass,  $s_* = \log \min \mathcal{P}$ , and  $G = \lfloor \log_2(\lambda/s_*) \rfloor + 1$ . Then every majorant  $\nu$  of  $A$  of



level  $\lambda \geq s_*$  satisfies, for every  $\alpha > 0$ ,

$$\begin{aligned} \log \frac{1}{\mathbb{E}\nu} &\leq \log \frac{Q_0}{|R|} + 19\alpha\lambda + C_4 \sum_{\ell \in \mathcal{P}, \ell \leq e^\lambda} \bar{p}_\ell \ell^{-\alpha} \\ &\quad + G(75 + \log(2 + \lambda/s_*)) + \frac{G}{2} \log(16\bar{\mu} + 16). \end{aligned} \quad (4)$$

If  $\lambda < s_*$ ,  $\nu$  depends only on the small residue and the bound holds without the  $\lambda$ -terms.

*Proof.* Use the uniform measure on  $\mathbb{Z}/Q_{\text{tot}}$  for a common period. By CRT the coordinates  $n \bmod Q_0^*$  (the  $Q_0$ -primary part of  $Q_{\text{tot}}$ ),  $n \bmod \ell^{e_\ell}$  ( $\ell \in \mathcal{P}$ ) and the rest are independent and uniform. Since  $\nu \geq 0$ ,  $\mathbb{E}\nu \geq (|R|/Q_0) \text{avg}_{c \in R} \mathbb{E}[\nu \mid n \bmod Q_0 = c]$ . Fix  $c \in R$ . The hit indicators  $x_\ell = \mathbf{1}[n \bmod \ell \in F_\ell(c)]$  are independent Bern( $p_\ell(c)$ ), independent of the finer digits. Each term of  $\nu$ , conditioned on  $(c, x)$ , depends only on  $x_{T_i}$  with  $T_i = \{\ell \in \mathcal{P} : \ell \mid d_i\}$ , so  $\nu_c = \mathbb{E}[\nu \mid c, x]$  is  $\lambda$ -level with weights  $s_\ell = \log \ell$ , and  $\nu_c \geq 0$ . The event  $\{c\} \times \{x = 0\}$  has positive probability and lies in  $\mathcal{A}$ , so  $\nu_c(0) \geq 1$ . Apply Proposition 4.4 in each fibre, then Jensen over  $c \in R$ ; the right side of (3) is linear in the  $p_\ell$  except for the concave logarithmic term.  $\square$

*Remark 4.6* (size of the error terms). The last two terms of (4) are  $O(\log^2 \lambda)$  when  $s_* \gg 1$  and  $\log \bar{\mu} \ll \log \lambda$ . In general  $\bar{\mu}$  may be as large as  $\pi(e^\lambda)/4$ , and then they are  $\asymp \lambda \log \lambda$ . In every application below  $\bar{\mu} \ll \lambda^3$ .

*Remark 4.7* (the  $R$ -term). The term  $\log(Q_0/|R|)$  is a saving that the theorem leaves uncontrolled:  $\nu = \mathbf{1}_R$  is a majorant with no slice cost and saving exactly  $\log(Q_0/|R|)$ . For a selector  $R = \{(c, P) = 1\}$  the term is  $\log(P/\varphi(P)) \leq \log \log \log P + O(1)$ . If  $R$  itself encodes forced classes with moduli dividing  $Q_0$ , the term is the void probability of a non-slice subsystem, about which Theorem 4.5 says nothing.

Define the *Rankin functional*

$$\Psi(\lambda) = \inf_{\alpha > 0} \left[ \alpha\lambda + \sum_{\ell} \bar{p}_\ell \ell^{-\alpha} \right]. \quad (5)$$

Theorem 4.5 says that no majorant of level  $\lambda$  saves more than  $C\Psi(\lambda) + O(\log^2 \lambda)$  beyond the  $R$ -term, in the situation of Remark 4.6. No converse is claimed in general; attainability is discussed only for the two campaign families (Section 10).

*Remark 4.8* (Montgomery's large sieve; [ET, Remark 2.6]; [PROVED]). Montgomery's large sieve [M] applied in a fibre  $c$  gives a bound  $Y/S_c(Q)$  with  $Q^2 \leq Y$ , where  $S_c(Q) = \sum_{q \leq Q} \mu^2(q) \prod_{\ell \mid q} g_\ell$  and  $g_\ell = |F_\ell(c)|/(\ell - |F_\ell(c)|)$ . Rankin's trick gives  $S_c(Q) \leq Q^\alpha \prod_{\ell} (1 + g_\ell \ell^{-\alpha}) \leq \exp\{\alpha \log Q + 2 \sum_{\ell} p_\ell(c) \ell^{-\alpha}\}$  when  $p_\ell(c) \leq 1/2$ . Summed over fibres, Jensen gives  $\sum_c Y e^{-\Phi_c} \geq |R| Y e^{-\text{avg } \Phi_c}$ . So the large-sieve bound over a slice system obeys the same functional, with level  $\lambda = \log Q$ , directly and without duality. No sequential or shared-prime large-sieve statement is made. The condition  $p_\ell(c) \leq 1/2$ , which gives  $g_\ell \leq 2p_\ell(c)$ , is the reading of the Rankin step in [ETr, item 3.4].

**4.4. Sequential windows: shared large primes.** Theorem 4.5 needs the slice primes to be disjoint from the small modulus. The following extension allows a large prime to be the slice prime of one condition and a cofactor prime of another.



Setting [ET, §2.6]. Take  $Q_0, R \subseteq \mathbb{Z}/Q_0$ , and a set  $\mathcal{P}$  of primes not dividing  $Q_0$ , split into ordered windows  $\mathcal{P} = W_1 \sqcup \cdots \sqcup W_J$ . A *condition*  $C$  is an event

$$E_C = \{n \equiv a_C \pmod{q_C}, n \equiv b_{C,\ell} \pmod{\ell^{e_{C,\ell}}} \text{ for } \ell \in S(C)\}$$

with  $q_C \mid Q_0$ ,  $\emptyset \neq S(C) \subseteq \mathcal{P}$  and  $e_{C,\ell} \geq 1$ . Assume

- (U) the last window meeting  $S(C)$  meets it in exactly one prime  $\ell(C)$ , and  $e_{C,\ell(C)} = 1$ .

Fix  $E_\ell \geq \max_C e_{C,\ell}$ . The *history* before window  $j$  is  $H_{<j} = (n \bmod Q_0, (n \bmod \ell^{E_\ell})_{\ell \in W_{<j}})$ . For  $\ell \in W_j$  and a history  $h$  let  $F_\ell(h)$  be the set of  $b_{C,\ell}$  over the conditions  $C$  with  $\ell(C) = \ell$  whose other requirements are met by  $h$ , and  $p_\ell(h) = |F_\ell(h)|/\ell$ . The avoider set is  $\mathcal{A} = \{n : n \bmod Q_0 \in R, n \notin E_C \forall C\}$ . Let  $Q_{\text{seq}}$  be the law of the history built window by window:  $c = n \bmod Q_0$  uniform on  $R$ ; given  $H_{<j}$ , the residues  $n \bmod \ell$  ( $\ell \in W_j$ ) independent and uniform on  $\mathbb{Z}/\ell \setminus F_\ell(H_{<j})$ , with uniform independent higher digits. Majorants of level  $\lambda$  charge all primes of  $\mathcal{P}$ .

**Theorem 4.9** (sequential sieve limit; [ET, Thm 2.7]; [PROVED]). *Assume that on the support of  $Q_{\text{seq}}$ ,  $p_\ell(h) \leq 1/4$  whenever  $\log \ell \leq \lambda$ , and  $p_\ell(h) < 1$  always. Then for all  $\alpha_1, \dots, \alpha_J > 0$ , every majorant  $\nu$  of level  $\lambda$  satisfies*

$$\log \frac{1}{\mathbb{E}\nu} \leq \log \frac{Q_0}{|R|} + \sum_j \mathbb{E}_{Q_{\text{seq}}} [\Phi_j(H_{<j})],$$

where  $\Phi_j(h)$  is the right side of (3) for the coordinates  $\ell \in W_j$  with  $\log \ell \leq \lambda$ , probabilities  $p_\ell(h)$  and  $\alpha = \alpha_j$ . A window with  $\log \min W_j > \lambda$  contributes 0.

*Proof.* Let  $g_j(h) = \mathbb{E}[\nu \mid H_{<j} = h]$  under the uniform measure, and let  $\mathcal{A}_{<j}$  be the histories in  $R$  satisfying no condition whose last window is before  $j$ . We show by downward induction that for  $h \in \mathcal{A}_{<j}$ ,  $-\log g_j(h) \leq \mathbb{E}_{Q_{\text{seq}}} [\sum_{i \geq j} \Phi_i(H_{<i}) \mid H_{<j} = h]$ . For  $j = J+1$  the history decides membership in  $\mathcal{A}$ , and  $\nu \geq 1$  on  $\mathcal{A}$ . For the step, fix  $h \in \mathcal{A}_{<j}$ . Given  $h$ , the residues  $n \bmod \ell^{E_\ell}$ ,  $\ell \in W_j$ , are independent and uniform, so the indicators  $x_\ell = \mathbf{1}[n \bmod \ell \in F_\ell(h)]$  are independent  $\text{Bern}(p_\ell(h))$ . Put  $f(x) = \mathbb{E}[g_{j+1}(H_{<j+1}) \mid h, x]$ . Then  $\mathbb{E}f = g_j(h)$ ,  $f \geq 0$ , and  $f$  is  $\lambda$ -level on  $W_j$ . The event  $x = 0$  is exactly that no condition with last window  $j$  holds; given it,  $H_{<j+1} \in \mathcal{A}_{<j+1}$  with the  $Q_{\text{seq}}$  transition law. The induction hypothesis and Jensen give  $f(0) > 0$  and the bound for  $f(0)$ ; Proposition 4.4 for  $f/f(0)$  gives  $\mathbb{E}f \geq f(0)e^{-\Phi_j(h)}$ . Finally  $\mathbb{E}\nu \geq (|R|/Q_0) \text{avg}_{c \in R} g_1(c)$  and Jensen over  $c$ .  $\square$

**Lemma 4.10** (profile under  $Q_{\text{seq}}$ ; [ET, Lemma 2.8]; [PROVED]). *For  $\ell \in W_j$ ,*

$$\mathbb{E}_{Q_{\text{seq}}} p_\ell(H_{<j}) \leq \sum_{C: \ell(C)=\ell} \frac{1}{\ell} \mathbb{P}(c \equiv a_C \pmod{q_C} \mid c \in R) \prod_{\ell' \in S(C) \setminus \{\ell\}} \frac{1}{\ell'^{e_{C,\ell'}} (1 - p_{\ell'}^*)},$$

where  $p_{\ell'}^*$  is the supremum of  $p_{\ell'}(h)$  over reachable histories.

*Proof.* Chain rule over windows: given the past,  $n \bmod \ell'$  is uniform on at least  $\ell'(1 - p_{\ell'}^*)$  residues, and the higher digits are uniform.  $\square$

## 5. SUPPLY PROFILES

We use Shiu's theorem [S, Theorem 1] in the form stated in [TQ, §3]: if  $F \geq 0$  is multiplicative,  $F(p^a) \leq A_1^a$ , and  $F(n) \ll_\epsilon n^\epsilon$ , then for fixed  $0 < \alpha_s, \beta_s < 1/2$ ,

$$\sum_{\substack{V-Y < n \leq V \\ n \equiv a \pmod{q}}} F(n) \ll \frac{Y}{\varphi(q) \log V} \exp \left\{ \sum_{p \leq V, p \nmid q} \frac{F(p)}{p} \right\}$$

uniformly for  $q \geq 2$ ,  $(a, q) = 1$ ,  $q < Y^{1-\alpha_s}$ ,  $V^{\beta_s} < Y \leq V$ .

**Lemma 5.1** (cubic Case-B profile,  $\mathcal{R}(M)$ ; [ET, Lemma 3.1]; [PROVED]). *Put  $S_B(x) = \sum_{M \leq x, M \equiv 3 \pmod{4}} \tau(A^2)M/\varphi(M)$  with  $A = (M+1)/4$ . Then  $S_B(x) \ll x(\log 2x)^2$ . Consequently, for  $0 < \beta \leq 1$ ,  $\sum_{M \equiv 3 \pmod{4}} \tau(A^2)(M/\varphi(M))M^{-1-\beta} \leq C_6\beta^{-3}$ .*

*Proof.* Write  $M/\varphi(M) = \sum_{d|M} \mu^2(d)/\varphi(d)$  and swap sums; the inner sum runs over  $A \leq (x+1)/4$  with  $4A \equiv 1 \pmod{d}$ . For  $2 \leq d \leq x^{1/2}$ , Shiu with  $F(A) = \tau(A^2)$  ( $F(p) = 3$ ) in the reduced class  $4^{-1} \pmod{d}$  gives  $\ll (x/\varphi(d))(\log x)^2$ ; for  $d = 1$  the elementary bound  $\sum_{A \leq x} \tau(A^2) \ll x \log^2 x$  suffices; and  $\sum_d \varphi(d)^{-2} < \infty$ . For  $d > x^{1/2}$  bound trivially by  $(x/d+1)x^{1/10}$ , for a total  $\ll x^{0.61}$ . The  $\beta$ -bound follows by partial summation,  $\int (\log 2x)^2 x^{-1-\beta} dx \ll \beta^{-3}$ .  $\square$

Numerically  $S_B(x)/(x \log^2 x) = 0.1067, 0.0990, 0.0963$  at  $x = 10^2, 10^4, 10^6$  [EVIDENCE].

**Lemma 5.2** ( $((a, D)$  profile; [ET, Lemma 3.2]; [PROVED]). *With  $G = 4a g(D)$ ,  $\sum_{a,D} (G/\varphi(G))G^{-1-\beta} \ll \beta^{-3}$  for  $0 < \beta \leq 1$ .*

*Proof.* Using  $\#\{D : g(D) = g\} \leq 2^{\omega(g)}$ , the sum is at most

$$\ll \left( \sum_a a^{-1-\beta} \frac{a}{\varphi(a)} \right) \left( \sum_g 2^{\omega(g)} \frac{g}{\varphi(g)} g^{-1-\beta} \right) \ll \beta^{-1} \beta^{-2}.$$

$\square$

**Lemma 5.3** (weighted Case-A divisor sum; [ET, Lemma 3.7]; [PROVED MOD] Elsholtz–Tao Prop. 1.4). *For  $x \geq 2$ ,*

$$\sum_{r,h \geq 1, rh \leq x} \tau(4rh^2 + 1) \frac{rh}{\varphi(rh)} \ll x(\log x)^2. \quad (6)$$

*The same holds with the extra weight  $\gamma(rh)$  of Corollary 6.2 below. Consequently the Case-A classes  $-m^{-1} \pmod{4g(d)}$ , with the selector weight, have profile  $\ll \beta^{-3}$ .*

*Proof.* The input is [EIT, Prop. 1.4]: for  $A, B > 1$  and every positive integer  $k \ll (AB)^{O(1)}$ ,  $\sum_{a \leq A} \sum_{b \leq B} \tau(kab^2 + 1) \ll AB \log(A+B) \log(1+k)$ . Use  $n/\varphi(n) \leq (\pi^2/6) \sum_{d|n} 1/d$  and write  $r = sr'$ ,  $h = th'$ , so that  $4rh^2 = (4st^2)r'h'^2$ . The left side of (6) is at most  $C \sum_{s,t} (st)^{-1} \sum_{r'h' \leq Y} \tau(4st^2r'h'^2 + 1)$  with  $Y = x/(st)$ . If  $st \leq \sqrt{x}$ , cover  $r'h' \leq Y$  by  $O(\log Y)$  rectangles with  $AB \asymp Y \geq \sqrt{x}/2$ ,  $A, B \geq 2$ ; then  $k = 4st^2 \ll (AB)^3$ , and Prop. 1.4 gives  $\ll Y(\log x)^2 \log(1 + 4st^2)$ ; finally  $\sum_{s,t} \log(1 + 4st^2)/(st)^2 < \infty$ . If  $st > \sqrt{x}$  then  $Y < \sqrt{x}$  and  $\tau \ll x^\epsilon$  give  $\ll x^{1/2+2\epsilon}$ . For the  $\gamma$ -weight replace  $1/(st)$  by  $\eta(s)\eta(t)$ , where  $\gamma(n)n/\varphi(n) = \sum_{d|n} \eta(d)$  and  $\eta(p) = O(p^{-1} + p^{-\delta})$ ; the needed sums converge because  $\eta(p)/p \ll p^{-1-\delta}$  and

$\sum_n \eta(n) n^{-1+\delta/2} < \infty$ . The profile statement follows by partial summation, since the Case-A class mass up to modulus  $G$  is at most  $\sum_{4rh \leq G} \tau(4rh^2 + 1)/(4rh)$ .  $\square$

The review [ETr, item 4] checked Prop. 1.4 and its  $k$ -range against the source; it was not re-proved.

## 6. DOMINANT-PRIME FAMILIES

### 6.1. Prime-slice families.

**Corollary 6.1** ([ET, Cor. 3.4]; [PROVED]; Case A [PROVED MOD] Elsholtz–Tao Prop. 1.4). *Fix  $0 < C < 1$  and  $A \geq 1$ . Take a prime-slice system in which every condition is a slice condition (no pure small-modulus conditions), and*

- *every condition is a forced class of  $\mathcal{R}(M)$  (any splitting  $M = q_0 \ell$ ), an  $(a, D)$ -class, or a Case-A class;*
- *each condition modulus  $q_0 \ell$  satisfies  $q_0 \leq \ell^C$ ;*
- *$R = \{c : (c, P) = 1\}$  for some  $P \mid Q_0$ ;*
- *$\ell \geq \ell_0(C)$  for every slice prime.*

*Then every majorant of level  $\lambda \leq A \log N$  satisfies*

$$\log \frac{1}{\mathbb{E}_\nu} \leq C_7(A, C)(\log N)^{3/4} + \log \frac{P}{\varphi(P)},$$

*and so does the large-sieve bound of Remark 4.8, summed over fibres.*

*Proof.* Since  $q_0 \leq \ell^C$  and the number of family classes modulo  $M$  is  $M^{o(1)}$ ,  $|F_\ell(c)| \leq \ell^{C+o(1)} \leq \ell/4$ . For a selector,  $\mathbb{P}(c \equiv b \pmod{q_0} \mid R) \leq 1/\varphi(q_0)$ , so  $\bar{p}_\ell \leq \sum_{M=q_0 \ell} |\mathcal{C}(M)|(M/\varphi(M))/M$ , where  $|\mathcal{C}(M)|$  is the number of family classes modulo  $M$ . Since  $\ell \geq M^{1/(1+C)}$ ,  $\ell^{-\alpha} \leq M^{-\alpha/2}$ . Lemmas 5.1, 5.2 and 5.3 give  $\sum_\ell \bar{p}_\ell \ell^{-\alpha} \ll \alpha^{-3}$ . Take  $\alpha = \lambda^{-1/4}$  in (4); here  $G = O(\log \lambda)$  and  $\bar{\mu} \ll \lambda^3$ .  $\square$

### 6.2. Dominant-prime families.

**Corollary 6.2** ([ET, Cor. 3.6] and §3.7; [PROVED]; for Case-A classes [PROVED MOD] Elsholtz–Tao Prop. 1.4). *Fix  $0 < C < 1$  and  $A \geq 1$ , and put  $\delta = (1-C)/2$ . Consider any family of Case-B forced classes (classes of  $\mathcal{R}(M)$ ), or  $(a, D)$ -classes modulo  $G = 4a g(D)$ ), and Case-A classes (modulo  $G = 4rh$ ), such that every modulus  $M$  has a prime factor  $\ell(M) \geq M^{1/(1+C)}$ . The other prime factors are unrestricted: they may be large, occur to higher powers, or be the dominant prime of other moduli. Let  $w_0 = w_0(C)$  be large,  $Q_0 = \text{lcm}(P_{w_0}, w_0\text{-smooth parts of all moduli})$  and  $R = \{c : (c, P_{w_0}) = 1, c \text{ satisfies no condition with all primes } \leq w_0\}$ . Then every majorant of level  $\lambda \leq A \log N$ , charging all primes  $> w_0$ , satisfies*

$$\log \frac{1}{\mathbb{E}_\nu} \leq C_8(A, C)(\log N)^{3/4} + \log \frac{Q_0}{|R|}, \quad \log \frac{Q_0}{|R|} = O_C(1).$$

*Proof. The  $R$ -term.* A  $w_0$ -smooth modulus is at most  $w_0^{1+C}$ , so there are finitely many. Let  $L' = \text{lcm}(P_{w_0}, w_0\text{-smooth moduli})$ . By Lemma 2.5,  $c \equiv 1 \pmod{L'}$  lies in  $R$ , so  $|R|/Q_0 \geq 1/L' = e^{-O_C(1)}$ .

*Windows.* With  $s_0 = \log w_0$  take  $W_j = (e^{s_j}, e^{s_j/C}]$ ,  $s_j = s_0 C^{-j+1}$ , and attach each condition to  $\ell(M)$ . Since  $\ell(M) > \sqrt{M}$  it occurs to exponent one, and every other prime  $\ell' \mid M$  has  $\ell' \leq M/\ell \leq \ell^C$ , so it lies in an earlier window or divides  $Q_0$ . Thus (U) holds.

*Hypotheses of Theorem 4.9.* The number of conditions with dominant prime  $\ell$  is at most  $\sum_{q \leq \ell^C} \ell^{o(1)} = \ell^{C+o(1)}$ , so  $p_\ell^* \leq \ell^{-\delta} \leq 1/4$  for  $\ell > w_0$ .

*Profile.* By Lemma 2.5,  $\mathbb{P}(c \equiv a \pmod{q} \mid R) \leq L'/\varphi(q)$ . By Lemma 4.10 each condition contributes at most  $L'(M/\varphi(M))\gamma(M)/M$  with  $\gamma(M) = \prod_{\ell' \mid M, \ell' > w_0} (1 - \ell'^{-\delta})^{-1}$ . Write  $(M/\varphi(M))\gamma(M) = \sum_{d \mid M} h(d)$ ,  $h$  multiplicative on squarefree  $d$  with  $h(p) \leq 1/(p-1) + 3p^{-\delta}$ . The proof of Lemma 5.1 goes through: in the Shiu range it uses  $\sum_d h(d)/\varphi(d) < \infty$ ; for  $d > x^{1/2}$ ,  $\sum_d h(d)d^{-1+\delta/2} < \infty$  gives a tail  $\ll x^{1-\delta/4+\epsilon} = o(x)$ . For  $(a, D)$ -classes use  $(G/\varphi(G))\gamma(G) \leq 2[(a/\varphi(a))\gamma(a)][(g/\varphi(g))\gamma(g)]$ , and for Case A use the  $\gamma$ -weighted form of Lemma 5.3. So the profile is  $\ll_C \beta^{-3}$ .

*Summing windows.* For  $\ell \in W_j$ ,  $M \leq \ell^{1+C} \leq e^{2s_j/C}$ , so the window mass is  $\ll s_j^3$  and its profile is  $\ll e^{-\alpha s_j} s_j^3$ . Optimising  $\alpha_j$  per window,  $\mathbb{E}\Phi_j \ll \min\{s_j^3, (\lambda/s_j)(1 + \log^+(s_j^4/\lambda))\} + O(\log^2 \lambda)$ . Only  $J \ll_C \log \lambda$  windows meet  $\ell \leq e^\lambda$ . Below the crossover  $s \approx \lambda^{1/4}$  the cubic branch is a geometric sum; above it, with  $s = \lambda^{1/4} C^{-k}$ , the window contributes  $O_C(\lambda^{3/4} C^k(1+k))$ ; the remainders total  $O_C(\log^3 \lambda)$ . Hence  $\sum_j \mathbb{E}\Phi_j \ll_C \lambda^{3/4}$ .  $\square$

The constants deteriorate as  $C \rightarrow 1$ . For unweighted integers, Dickman's function gives logarithmic density  $\log(1+C)$  to the moduli with  $P(M) \geq M^{1/(1+C)}$ ; the share of the *weighted* cubic supply is not proved [CONJECTURE].

## 7. THE QUADRATIC-RESIDUE BASE AND THE CAPPED MEASURE

Corollary 6.2 needs  $P(M) \geq M^{1/(1+C)}$  with  $C < 1$ . This section gives the two tools that remove this restriction for the classes  $\mathcal{R}(M)$ : a product base measure on the avoiders of all small-modulus classes (Lemma 7.2), and a capped sequential measure whose heavy coordinates enter only through a leak (Theorem 7.4). With them it proves the cap for gapped and window-resolved families (Theorems 7.6 and 7.7). These two theorems are now special cases of Theorem 9.3; they are kept because their proofs need only the Boolean step of Proposition 4.4. The results are from [TW, §§1–2, 4], building on the reductions of [EB, §2]; the review [TWr] (two rounds) found them sound.

**7.1. Gapped and twin moduli.** Write  $P_2(M) = P(M/P(M))$ , with  $P_2(1) = 1$ . Fix  $w_0 \geq 3$ ,  $0 < \eta < 1$  and  $B \geq 0$ . Following [EB, §2],  $M$  is  $\eta$ -gapped if  $P(M) > w_0$  and  $P(M) \geq P_2(M)^{1+\eta}$ , and  $\eta$ -twin otherwise; it is  $(\eta, B)$ -gapped if moreover  $M \leq P(M)^{1+B}$ . Put  $s_0 = \log w_0$ ,  $s_j = s_0(1+\eta)^{j-1}$  and use the windows  $W_j = \{\ell \text{ prime} : s_j < \log \ell \leq s_{j+1}\}$ .

**Lemma 7.1** (gapped implies (U); [EB, Lemma 2.1]; [PROVED]). *If  $M$  is  $\eta$ -gapped and  $\ell = P(M)$ , then  $\ell \parallel M$ , and every prime  $\ell' \mid M/\ell$  with  $\ell' > w_0$  lies in a window strictly before that of  $\ell$ . So a family of  $\eta$ -gapped classes satisfies (U) with  $\ell(C) = P(M_C)$ .*

*Proof.* If  $\ell^2 \mid M$  then  $P_2(M) = \ell$ , contradicting  $\ell \geq \ell^{1+\eta}$ . If  $\ell' \in W_j$  divides  $M/\ell$ , then  $\log \ell \geq (1+\eta) \log \ell' > (1+\eta)s_j = s_{j+1}$ .  $\square$

Dominant moduli with  $P(M) > w_0$  are  $\eta$ -gapped for every  $\eta < 1$  with  $\eta \leq 1/C - 1$ . Balanced moduli  $M = k\ell_1\ell_2$  with  $P(k) < \ell_1$ ,  $\ell_2 \geq \ell_1^{1+\eta}$  and  $k\ell_1 \geq \ell_2$  are  $\eta$ -gapped. The same-scale pairs of Lemma 13.1 below are  $\eta$ -twin. For a gapped balanced modulus the cofactor  $M/P(M)$  exceeds  $P(M)$ , so the counting bound  $p_\ell \leq \ell^{C-1+o(1)}$

of Corollary 6.2 is lost; the hit probabilities can be large for some histories (*heavy coordinates*).

**7.2. The quadratic-residue base.** Fix  $W \geq 3$ , let  $Q_0$  be a common multiple of the  $W$ -smooth parts of all moduli with  $P_W \mid Q_0$ , and put  $R_W = \{c \bmod Q_0 : (\frac{c}{p}) = 1 \text{ for every odd prime } p \leq W\}$ .

**Lemma 7.2** (QR base; [TW, Lemma 1.3]; [PROVED]). (1) Every  $c \in R_W$  avoids every class of  $\mathcal{R}(M)$  with  $W$ -smooth  $M$ . (2)  $\log(Q_0/|R_W|) = \sum_{3 \leq p \leq W} \log(2p/(p-1))$ . (3) Under the uniform law on  $R_W$ , residues at distinct primes are independent and  $\mathbb{P}(c \equiv a \pmod{p^e}) \leq \gamma(p)/p^e$  for  $p^e \mid Q_0$ , with  $\gamma(p) = 2p/(p-1)$  for odd  $p$  and  $\gamma(2) = 1$ .

*Proof.* (1) is Lemma 2.6. (2) and (3): by CRT  $R_W$  is a product over the prime powers  $p^e \parallel Q_0$ ; for odd  $p \leq W$  the factor has  $p^{e-1}(p-1)/2$  elements out of  $p^e$ .  $\square$

So Mordell's obstruction, which classically shows that polynomial identities never cover squares, is used constructively: it gives a *product* base measure on avoiders of all small-modulus classes, with polylogarithmic inflation.

**7.3. The capped sequential measure.** Work in the sequential setting of Section 4.4, with  $R$  satisfying (R1) every  $c \in R$  avoids every condition with  $S(C) = \emptyset$ , and (R2) under the uniform law on  $R$  the residues at distinct primes are independent and  $\mathbb{P}(c \equiv a \pmod{p^e}) \leq \gamma(p)/p^e$ . Fix caps  $\delta_\ell \in (0, 1/4]$ . The *capped measure*  $Q'$  draws  $c$  uniformly from  $R$ ; given  $H_{<j} = h$ , the residues  $n \bmod \ell$  ( $\ell \in W_j$ ) are independent, uniform on  $\mathbb{Z}/\ell \setminus F_\ell(h)$  if  $p_\ell(h) \leq \delta_\ell$  (*light*), and uniform on  $\mathbb{Z}/\ell$  otherwise (*heavy*); higher digits are uniform. The *leak* is  $\mathfrak{L} = Q'(\text{final history} \notin \mathcal{A})$ . This is a variant of Hough's distortion method [H], in the form of Balister–Bollobás–Morris–Sahasrabudhe–Tiba [BBMST], which bounds the density of the uncovered set (the majorant  $\mathbf{1}_\mathcal{A}$  only), adapted here to general level- $\lambda$  majorants.

**Lemma 7.3** (leak and inflation; [TW, Lemmas 2.1, 2.2]; [PROVED]). (a)  $\mathfrak{L} \leq \sum_{\ell \in \mathcal{P}} \mathbb{E}_{Q'}[p_\ell(H_{<j(\ell)}) \mathbf{1}\{p_\ell(H_{<j(\ell)}) > \delta_\ell\}]$ . (b) For  $m$  a product of prime powers  $p^e$  with  $e \leq E_p$  ( $p \in \mathcal{P}$ ) or  $p^e \mid Q_0$ , and every  $a$ ,  $Q'(n \equiv a \pmod{m}) \leq \prod_{p^e \parallel m} \gamma'(p)/p^e$ , where  $\gamma'(p) = \gamma(p)$  for  $p \mid Q_0$  and  $\gamma'(p) = (1 - \delta_p)^{-1}$  for  $p \in \mathcal{P}$ .

*Proof.* (a) A final history outside  $\mathcal{A}$  satisfies some condition  $C$  with  $S(C) \neq \emptyset$ , by (R1). At the transition into the window of  $\ell = \ell(C)$ ,  $b_{C,\ell} \in F_\ell(h)$ , and light coordinates never take such values, so  $\ell$  was heavy and the hit has probability  $p_\ell(h)$ ; take a union bound. (b) Chain rule: (R2) at the base; a light coordinate is uniform on at least  $p(1 - \delta_p)$  residues, a heavy one on all  $p$ .  $\square$

**Theorem 7.4** (capped sequential sieve limit; [TW, Thm 2.3]; [PROVED]). If  $\mathfrak{L} \leq 1/2$ , every majorant  $\nu$  of level  $\lambda$  of  $\mathcal{A}$  satisfies

$$\log \frac{1}{\mathbb{E}\nu} \leq \log \frac{Q_0}{|R|} + \log 2 + 2 \sum_j \mathbb{E}_{Q'}[\Phi_j^{\text{light}}(H_{<j})],$$

where  $\Phi_j^{\text{light}}(h)$  is the right side of (3) for the coordinates of  $W_j$  that are light at  $h$  and have  $\log \ell \leq \lambda$ .

*Proof.* Let  $g_j(h) = \mathbb{E}_U[\nu \mid H_{<j} = h]$  under the uniform measure. By downward induction, for every history  $h$  at window  $j$ ,

$$g_j(h) \geq \mathbb{E}_{Q'} \left[ \mathbf{1}_{\mathcal{A}}(H_{<J+1}) \exp \left( - \sum_{i \geq j} \Phi_i^{\text{light}}(H_{<i}) \right) \mid H_{<j} = h \right].$$

The base case is  $g_{J+1} \geq \mathbf{1}_{\mathcal{A}}$ . For the step, the hit indicators under  $U$  given  $h$  are independent  $\text{Bern}(p_\ell(h))$ ; split  $x = (x_L, x_H)$  into light and heavy and put  $\tilde{f}(x_L) = \mathbb{E}_U[g_{j+1}(H_{<j+1}) \mid h, x_L]$ . Then  $\mathbb{E}\tilde{f} = g_j(h)$ ,  $\tilde{f} \geq 0$ ,  $\tilde{f}$  is  $\lambda$ -level in  $x_L$  (averaging over  $x_H$  preserves this), light coordinates have  $p_\ell(h) \leq 1/4$ , and given  $x_L = 0$  the law of  $H_{<j+1}$  is the  $Q'$  transition. If  $\tilde{f}(0) = 0$  the claim is trivial; otherwise Proposition 4.4 for  $\tilde{f}/\tilde{f}(0)$  gives  $\mathbb{E}\tilde{f} \geq \tilde{f}(0)e^{-\Phi_j^{\text{light}}(h)}$ , and the induction hypothesis bounds  $\tilde{f}(0)$ . Finally, with  $S = \sum_j \Phi_j^{\text{light}} \geq 0$ ,  $\mathbb{E}\nu \geq (|R|/Q_0)\mathbb{E}_{Q'}[\mathbf{1}_{\mathcal{A}}e^{-S}]$ , and Jensen on  $Q'(\cdot \mid \mathcal{A})$  with  $Q'(\mathcal{A}) \geq 1/2$  gives  $\mathbb{E}_{Q'}[\mathbf{1}_{\mathcal{A}}e^{-S}] \geq \frac{1}{2}e^{-2\mathbb{E}_{Q'}S}$ .  $\square$

Heavy coordinates thus cost nothing in the bound; they enter only through  $\mathfrak{L}$ , which is controlled by an average, never by a supremum.

**7.4. The gapped cap.** Take the windows above with  $w_0 = W$ , the base  $R = R_W$ , and caps  $\delta_\ell = \ell^{-1/2}$ . The family  $\mathfrak{G}$  consists of  $\mathcal{R}(M)$ -classes with  $M$  ( $\eta, B$ )-gapped and  $P(M) > W$ , and any  $\mathcal{R}(M)$ -classes with  $W$ -smooth  $M$ . The latter are avoided by  $R_W$  (Lemma 7.2), so (R1) holds; (U) holds by Lemma 7.1. Then  $\Gamma(m) := \prod_{p|m} \gamma'(p) \leq 3^{\omega(m)}$ .

**Lemma 7.5** (second moment; [TW, Lemma 2.4]; [PROVED]). *For every  $\epsilon > 0$  there is  $C(\epsilon, B)$ , independent of  $W$ ,  $\eta$  and the family, such that  $\mathbb{E}_{Q'}[p_\ell(H)^2] \leq C(\epsilon, B)\ell^{-2+\epsilon}$  for every prime  $\ell > W$ .*

*Proof.*  $|F_\ell(h)|$  is at most the number  $N_\ell(n)$  of pairs  $(q, D)$  with  $q \leq \ell^B$ ,  $D \mid A_{q\ell}^2$  and  $n \equiv -4D \pmod{q}$ , a function of the full history. Expanding  $N_\ell^2$ , a compatible pair of pairs contributes  $Q'(n \equiv a \pmod{m}) \leq 3^{\omega(m)}/m$  with  $m = \text{lcm}(q, q')$ , by Lemma 7.3(b). Each  $q$  carries  $\tau(A^2) \ll_\epsilon \ell^{\epsilon/4}$  divisors  $D$ , and there are  $\tau(m^2)$  pairs with  $\text{lcm} = m$ . So  $\mathbb{E}N_\ell^2 \ll_\epsilon \ell^{\epsilon/2} \sum_{m \leq \ell^{2B}} 3^{\omega(m)}\tau(m^2)/m \ll_{\epsilon, B} \ell^\epsilon$ . Divide by  $\ell^2$ .  $\square$

By Lemma 7.3(a) and Markov,  $\mathfrak{L} \leq \sum_{\ell > W} \ell^{1/2}C(1/4, B)\ell^{-7/4} \ll_B W^{-1/4}$ , so there is  $W_0(B)$  with  $\mathfrak{L} \leq 1/2$  for  $W \geq W_0(B)$ , uniformly in  $\eta$  and the family [TW, Cor. 2.5]. The same weighted Shiu argument as in Corollary 6.2 (now with  $h(p) \leq 2$  for  $p \leq W$  and  $h(\ell) \leq 2\ell^{-1/2}$  for  $\ell > W$ ) gives, for every window with  $s_j \leq \lambda$ ,

$$\mathbb{E}_{Q'} \sum_{\ell \in W_j} p_\ell(H_{<j}) \leq K((1+B)s_j)^3, \quad K = K(W, B) \quad (7)$$

[TW, Lemma 2.6], using  $M \leq P(M)^{1+B} \leq e^{(1+B)s_{j+1}}$  [EB, Lemma 2.4].

**Theorem 7.6** (the gapped cap; [TW, Thm 2.7]; [PROVED]). *Fix  $B \geq 0$  and take  $W = W_0(B)$ . For every  $0 < \eta < 1$ , every family  $\mathfrak{G}$  as above, and every majorant  $\nu$  of level  $\lambda \geq \log W$ ,*

$$\log \frac{1}{\mathbb{E}\nu} \leq C K^{1/4}(1+B)^{3/4}\eta^{-1}\lambda^{3/4} + O_B(\eta^{-1}\log^2 \lambda) + O_B(1),$$

with  $C$  absolute and  $K = K(W_0(B), B)$ .

*Proof.* Theorem 7.4 applies since  $\mathfrak{L} \leq 1/2$ ; the  $R$ -term is  $O_B(1)$  by Lemma 7.2(2). Within a window the costs lie in  $(s_j, (1+\eta)s_j]$  and  $1+\eta < 2$ , so after Step 0 of Proposition 4.4 there is one nonempty band, and  $G = 1$  with  $s_*$  the least cost present [EB, Lemma 2.5a]. The light profile is at most the full profile, so (7) and concavity bound the logarithmic terms by  $O(\log \lambda + \log K + \log(1+B))$  per window, and there are at most  $1 + 2 \log \lambda / \eta$  windows with  $s_j \leq \lambda$ . The main term of window  $j$  is at most  $\inf_\alpha [19\alpha\lambda + X_j e^{-\alpha s_j}] \leq \min\{X_j, (19\lambda/s_j)(1 + \log^+(X_j s_j / 19\lambda))\}$  with  $X_j = C_4 K(1+B)^3 s_j^3$ . Put  $s^* = (19\lambda / (C_4 K(1+B)^3))^{1/4}$ . Windows with  $s_j \leq s^*$  contribute at most  $2X(s^*)/\eta = 38\lambda/(\eta s^*)$ ; the  $k$ -th window above  $s^*$  contributes at most  $(19\lambda/s^*)(1+\eta)^{-k}(1+4(k+1)\log(1+\eta))$ , summing to at most  $(19\lambda/s^*) \cdot 18/\eta$  [EB, proof of Thm 2.5]. The total is  $\leq 380\lambda/(\eta s^*)$ . Multiply by 2.  $\square$

For dominant moduli this recovers Corollary 6.2 for  $\mathcal{R}(M)$ -classes, with a different  $R$ -term. It needs no hypothesis on heavy coordinates: the extremal statement  $(E_\delta)$  and hypothesis  $H_{\text{light}}$  of [EB, §§2.5, 4.4] are not used.

*Scope of Theorem 7.6.* It covers only  $\mathcal{R}(M)$ -classes: Lemma 2.6 is proved for  $\mathcal{R}(M)$  only, and the analysis of [EB] is restricted to  $\mathcal{R}(M)$ . It needs  $B$  fixed: there is no summation over  $B$ . It does not cover  $\eta$ -twin moduli, for which (U) fails; Theorem 9.3 does.

**Theorem 7.7** (window-resolved families; [TW, Thm 4.4]; [PROVED]). *Fix  $B$ ,  $W = W_0(B)$  and  $\lambda \geq (2 \log W)^4$ . Let  $\mathfrak{G}$  be any family of  $\mathcal{R}(M)$ -classes with  $M \leq P(M)^{1+B}$ , plus any  $W$ -smooth classes, such that every modulus with top prime in  $(e^{\lambda^{1/4}}, e^{\lambda/2}]$  is window-resolved: the last  $\eta$ -window meeting its primes  $> W$  contains exactly one of them, to exponent one. Then every majorant of level  $\lambda$  satisfies  $\log(1/\mathbb{E}\nu) \ll_B \eta^{-1} \lambda^{3/4}$ .*

The proof [TW, §4] uses an abstract form of Theorem 7.4 with blocks: singletons below  $e^{\lambda^{1/4}}$ , where even twin and prime-power moduli cost  $\ll \lambda^{3/4}$  in total [TW, Prop. 4.1]; the  $\eta$ -windows in  $(e^{\lambda^{1/4}}, e^{\lambda/2}]$ ; one block above  $e^{\lambda/2}$ , where each term of  $\nu$  sees at most one prime and a linear-window inequality gives cost  $O(e^{-\lambda/4})$  [TW, Lemma 4.2, Cor. 4.3]; and a second-moment bound with prime-power top primes [TW, Lemma 4.0] for the leak. So all twin moduli with top prime outside  $(e^{\lambda^{1/4}}, e^{\lambda/2}]$  are already covered. The *residual* was the set of unresolved moduli with top prime in that range. Theorem 9.3 closes it by replacing the  $\eta$ -windows with dyadic sequential blocks and the Boolean step with Theorem 8.5; the other three ingredients are reused.

## 8. A WEIGHTED $k$ -ARY COMPARISON THEOREM

The sequential sieve limits of Sections 4.4 and 7 need, in each window, a single coordinate that decides each condition (hypothesis (U)). For twin moduli, prime-power tops, or several primes of one modulus at the same scale, (U) fails. This section proves an arithmetic-free comparison theorem of any arity that replaces the per-window Boolean step. Everything here is from [KA, §§1–3]; the reviews [KAr, KAr2] found it sound (the second re-derived all inputs; its presentational repairs E1, E2 are incorporated).

**8.1. The coupled sequential process.** Let  $V$  be a finite ordered set of coordinates,  $y_\ell \in \Omega_\ell$  finite, and  $\varpi = \bigotimes_\ell \varpi_\ell$  a product law. A *pattern* is a pair  $C = (T_C, a_C)$



with  $T_C \subseteq V$  and  $a_C \in \Omega_{T_C}$ ; put  $\text{top}(C) = \max T_C$ . Arities are arbitrary and unary patterns are allowed. A function  $f$  on  $\prod_\ell \Omega_\ell$  is  $d$ -local if  $f = \sum_T f_T$  with  $f_T$  a function of  $y_T$  and  $|T| \leq d$ ; a  $\lambda$ -level function on coordinates of cost  $> s$  is  $d$ -local with  $d = \lfloor \lambda/s \rfloor$ . Fix thresholds  $\delta_\ell \in (0, 1/4]$ .

The coordinates are processed in order. Before  $\ell$  the state is  $(c_i, y_i)_{i < \ell}$ , and the *plain activation rule* gives the set

$$F_\ell = F_\ell(y_{<\ell}) = \{a : \exists C, \text{top}(C) = \ell, (y_{T_C \setminus \{\ell\}}, a) = a_C\},$$

known before  $\ell$  is drawn; put  $p_\ell = \varpi_\ell(F_\ell)$ , and call  $\ell$  *light* if  $p_\ell \leq \delta_\ell$ . Draw  $c_\ell \sim \varpi_\ell$  independently of the past. If  $\ell$  is light and  $c_\ell \in F_\ell$  ( $\ell$  is *replaced*), draw  $y_\ell$  from  $\varpi_\ell(\cdot \mid \Omega_\ell \setminus F_\ell)$ , independently of  $c_\ell$ ; otherwise  $y_\ell = c_\ell$ . Write  $\omega = (c, y)$  for the path,  $J(\omega)$  for the replaced set,  $z(\omega) = |J(\omega)|$ , and  $m(\omega) = \sum_{\ell \text{ light}} p_\ell$  for the light activated mass. The law  $\sigma$  of  $y$  is the sequential capped law of [TW, §6]; the  $c$ 's are an explicit coupling of  $\sigma$  with  $\varpi$ .

**Lemma 8.1** (avoidance and inflation; [KA, Lemma 2.1]; [PROVED]). (1) If  $\text{top}(C)$  is light then  $y_{T_C} \neq a_C$ ; hence  $\mathbb{P}(y \text{ completes some } C) \leq \sum_\ell \mathbb{E}[p_\ell \mathbf{1}\{p_\ell > \delta_\ell\}]$ . (2) Given the past,  $y_\ell \sim \varpi_\ell(\cdot \mid \Omega_\ell \setminus F_\ell)$  if  $\ell$  is light and  $y_\ell \sim \varpi_\ell$  otherwise. Hence  $\sigma(y_S = b) \leq \prod_{\ell \in S} (1 - \delta_\ell)^{-1} \varpi(b)$  for all  $S \subseteq V$  and  $b \in \Omega_S$ .

*Proof.* (1) At a light  $\ell$ ,  $y_\ell \notin F_\ell$ , and every completion of a pattern with  $\text{top } \ell$  lies in  $F_\ell$ . At a heavy  $\ell$ ,  $y_\ell = c_\ell$  is independent of the past. (2) At a light  $\ell$ ,  $\mathbb{P}(y_\ell = a \mid \text{past}) = \varpi_\ell(a) \mathbf{1}_{a \notin F_\ell} (1 + p_\ell / (1 - p_\ell))$ ; the bound for  $y_S$  follows by the chain rule from the last element of  $S$  downwards.  $\square$

**8.2. Extrapolation in replacement coins.** Freeze a whole path  $\omega$ . For  $\rho \in \{0, 1\}^V$  put  $y_\ell^\rho = y_\ell$  if  $\ell \in J(\omega)$  and  $\rho_\ell = 1$ , and  $y_\ell^\rho = c_\ell$  otherwise. So  $y^0 = c$  and  $y^1 = y$ . The intermediate points  $y^\rho$  need not be paths of the process, and nothing below requires them to be.

**Lemma 8.2** (locality; [KA, Lemma 2.2]; [PROVED]). For fixed  $\omega$  and a  $d$ -local  $f \geq 0$ ,  $g_\omega(\rho) = f(y^\rho)$  is nonnegative, multilinear of degree at most  $d$  in  $\rho_{J(\omega)}$ , and independent of  $\rho_{V \setminus J(\omega)}$ .

*Proof.*  $y_\ell^\rho$  depends on  $\rho_\ell$  only, so  $f_T(y_T^\rho)$  is a function of  $\rho_{T \cap J}$ , and  $|T| \leq d$ .  $\square$

The one-dimensional input is Lagrange *extrapolation* for binomial laws: the node lemma (Lemma 4.2) interpolates at the end point 0, and here we extrapolate to the other end point  $z$ .

**Lemma 8.3** (binomial Lagrange extrapolation; [KA, Lemma 2.3 and §3]; [PROVED]). Let  $z \geq 0$ ,  $0 < t \leq 1/4$ ,  $d \geq 0$ , and let  $g \geq 0$  on  $\{0, 1\}^z$  be multilinear of degree at most  $d$ . Then  $g(1, \dots, 1) \leq B(z, t, d) \mathbb{E}_{\rho \sim \text{Bern}(t)^z} g(\rho)$ , where  $B(z, t, 0) = 1$  and, for  $d \geq 1$ ,

$$B(z, t, d) = \min_Y \max_{y \in Y} \frac{|\ell_y^Y(z)|}{\psi(y)}, \quad \psi = \text{Bin}(z, t),$$

over node sets  $Y \subseteq \{0, \dots, z\}$  with  $|Y| = \min(d, z) + 1$ ,  $\ell_y^Y$  the Lagrange basis of  $Y$ . One has  $B \geq 1$ , and

$$\log B(z, t, d) \leq d \log \left( C'_1 \max \left( \frac{1}{t}, \sqrt{\frac{z}{td}} \right) \right) + \frac{1}{2} \log(22zt + 22) + 3, \quad C'_1 = 2e^{4.31}. \quad (8)$$

*Proof.* Averaging  $g$  over  $\text{Sym}(z)$  changes neither side and gives  $Q(K)$ ,  $K = \sum \rho_i$ , with  $Q$  a polynomial of degree at most  $d$  (the average of  $\rho^S$  is  $\binom{K}{|S|}/\binom{z}{|S|}$ ) and  $Q \geq 0$  on  $\{0, \dots, z\}$ . Lagrange interpolation at a minimising  $Y$  gives  $Q(z) = \sum_y \ell_y(z)Q(y) \leq B \sum_y \psi(y)Q(y) \leq B \mathbb{E}Q(K)$ . For  $d \geq 1$ , interpolating the constant 1 gives  $\sum_y |\ell_y(z)| \geq 1 \geq \sum_y \psi(y)$ , so  $B \geq 1$ . For (8) use three node sets [KA, §3]:  $Y = \{0, \dots, z\}$  if  $z \leq d$  (then  $B \leq t^{-z}$ );  $Y = \{0, \dots, d\}$  if  $zt \leq 2d$ ; and, if  $zt > 2d$ ,  $d+1$  nodes spaced  $\lfloor \sqrt{zt/d} \rfloor$  around  $zt$ , bounded below by Lemma 4.1.  $\square$

The comparison with the product law rests on a supermartingale.

**Lemma 8.4** (thinned law; [KA, Lemma 2.4]; [PROVED]). *Fix  $t \in (0, 1/4]$  and put  $\mathcal{W}(\omega) = \exp(-\frac{4}{3}t m(\omega))$ . If  $\omega$  is drawn from the process and, given  $\omega$ ,  $\rho \sim \text{Bern}(t)^V$ , then for every  $x$ ,  $\mathbb{E}_\omega[\mathcal{W}(\omega)\mathbb{P}_\rho(y^\rho = x \mid \omega)] \leq \varpi(x)$ .*

*Proof.* At a replaced  $\ell$ ,  $c_\ell \in F_\ell \not\supseteq y_\ell$ , so  $c_\ell = x_\ell$  and  $y_\ell = x_\ell$  cannot both hold. Hence  $\mathbb{P}_\rho(y^\rho = x \mid \omega) \leq \prod_\ell \phi_\ell$  with  $\phi_\ell = \mathbf{1}\{c_\ell = x_\ell\} + t\mathbf{1}\{\ell \in J, y_\ell = x_\ell\}$ . Given the past,  $\mathbb{E}[\phi_\ell \mid \text{past}] = \varpi_\ell(x_\ell) + \mathbf{1}\{\ell \text{ light}\} t p_\ell \varpi_\ell(x_\ell) \mathbf{1}_{x_\ell \notin F_\ell} / (1 - p_\ell) \leq \varpi_\ell(x_\ell) D_\ell$ ,  $D_\ell = 1 + \frac{4}{3} t p_\ell \mathbf{1}\{\ell \text{ light}\}$ , and  $D_\ell$  is known before  $\ell$ . So the partial products of  $\phi_\ell / (\varpi_\ell(x_\ell) D_\ell)$  form a nonnegative supermartingale and  $Z = \prod_\ell \phi_\ell / (\varpi_\ell(x_\ell) D_\ell)$  has  $\mathbb{E}Z \leq 1$ . Pathwise  $\prod_\ell D_\ell \leq e^{\frac{4}{3}tm} = 1/\mathcal{W}$ , so  $\mathbb{E}[\mathcal{W} \prod \phi_\ell] = \varpi(x) \mathbb{E}[\mathcal{W} \prod D_\ell \cdot Z] \leq \varpi(x)$ .  $\square$

**Theorem 8.5** (weighted  $k$ -ary comparison; [KA, Thm 2.5]; [PROVED]). *In this setting, for every  $t \in (0, 1/4]$  and every  $d$ -local  $f \geq 0$ ,*

$$\mathbb{E}_\varpi f \geq \mathbb{E}_\omega [e^{-\Phi(\omega)} f(y)], \quad \Phi(\omega) = \log B(z(\omega), t, d) + \frac{4}{3} t m(\omega) \geq 0.$$

*No incident-weight, codegree or sparsity hypothesis is needed, and the arity of the patterns is arbitrary.*

*Proof.* By Lemmas 8.2 and 8.3,  $f(y) = g_\omega(\mathbf{1}_J) \leq B \mathbb{E}_\rho g_\omega(\rho)$ . Multiply by  $\mathcal{W}/B$  and take  $\mathbb{E}_\omega$ :  $\mathbb{E}_\omega [e^{-\Phi} f(y)] \leq \sum_x f(x) \mathbb{E}_\omega [\mathcal{W} \mathbb{P}_\rho(y^\rho = x \mid \omega)] \leq \mathbb{E}_\varpi f$  by Lemma 8.4 and  $f \geq 0$ .  $\square$

**Corollary 8.6** (mean cost; [KA, Cor. 2.6]; [PROVED]). *Let  $d \geq 1$ ,  $\bar{m} \geq \mathbb{E}m$  and  $t = d/(\bar{m} + 4d)$ . Then*

$$\mathbb{E}_\omega \Phi \leq d \log \frac{C_0(\bar{m} + 4d)}{d} + \frac{4}{3}d + \frac{1}{2} \log(22d + 22) + 3, \quad C_0 = 4e^{4.31}.$$

*Proof.* Each light  $\ell$  is replaced with conditional probability  $p_\ell$ , so  $\mathbb{E}z = \mathbb{E}m$ . Bound the maximum in (8) by the sum and use concavity of  $\log$  and  $\sqrt{\cdot}$ : with this  $t$ ,  $1/t = (\bar{m} + 4d)/d$ ,  $\sqrt{\mathbb{E}z/(td)} \leq (\bar{m} + 4d)/d$ ,  $t\mathbb{E}z \leq d$  and  $\frac{4}{3}t\mathbb{E}m \leq \frac{4}{3}d$ .  $\square$

So the step cost is  $d(O(1) + \log^+(\mathbb{E}m/d))$ , the form of Proposition 4.4 in one band, for patterns of any arity, and with only the *mean* activated mass.

**Remark 8.7** (weighted versus unweighted; [KA, Remark 2.8, §5]). Theorem 8.5 bounds  $\mathbb{E}[e^{-\Phi} f(y)]$ , not  $\log(\mathbb{E}_\sigma f / \mathbb{E}_\varpi f)$ . No bound of the latter by the mean mass holds in general [PROVED]: take a trigger  $X$  with  $\mathbb{P}(X = 1) = \epsilon$ , coordinates  $Z_1, \dots, Z_m$  with  $\mathbb{P}(Z_i = 1) = 1/4$ , patterns  $(X, Z_i) = (1, 1)$ ,  $\mu = m/4$  and  $\epsilon = 1/\mu$ ; every activated set is light,  $\mathbb{E}m(\omega) = 1$ , but the 3-local  $f = X(\sum Z_i - \mu)^2$  has  $\mathbb{E}_\sigma f / \mathbb{E}_\varpi f = \mu/(1 - 1/4) \rightarrow \infty$ . The incident weight of  $(X, 1)$  is about 1 here, so this does not contradict the unweighted comparison inequality conjectured in

[TW, Conj. 6.4] (Conjecture 14.2 below), which assumes small incident weights. That unweighted form remains [OPEN]; Theorem 8.5 is its weighted form, for the conjecture's own law  $\sigma$ . The sieve application below needs only the weighted form, because the sequential sieve limit accepts random step costs (Theorem 9.1). An exact check of Theorem 8.5 on 100 random small systems (exhaustive path enumeration and floating-point linear programmes) found no violation [EVIDENCE].

## 9. THE MAIN THEOREM: 3/4 FOR EVERY FIXED- $B$ FAMILY OF CLASSES $\mathcal{R}(M)$

This section combines Theorem 8.5 with the quadratic-residue base and the capped sequential measure of Section 7. The results are from [KA, §4]; the second review [KA2] re-derived every input taken from [TW] and found the argument sound.

**9.1. Sequential steps with random costs.** Take the sequential setting of Section 7.3: a base  $Q_0$ , an admissible set  $R$  with (R1)–(R2), coordinates  $y_\ell = n \bmod \ell^{E_\ell}$  for  $\ell \in \mathcal{P}$ , now grouped into ordered *blocks*  $V_1, \dots, V_J$  (in increasing order of primes) instead of windows. Each class is *decided* at its top prime  $P(M)$ : its other prime factors are smaller (earlier coordinates, or the base), and its requirement at  $P(M)$  is one class modulo  $P(M)^v$ . Hypothesis (U) is *not* assumed.

**Theorem 9.1** (weighted sequential sieve limit; [KA, Thm 4.1]; [PROVED]). *Suppose that for each block  $j$  and history  $h$  there are a probability law  $\Pi_j(h)$  on auxiliary paths  $\omega$ , a map  $\omega \mapsto Y_j(h, \omega) \in \prod_{\ell \in V_j} \mathbb{Z}/\ell^{E_\ell}$ , and a cost  $\Phi_j(h, \omega) \geq 0$ , such that*

$$(S_w) \text{ every } \lambda\text{-level } f \geq 0 \text{ on the block coordinates satisfies } \mathbb{E}_U f \geq \mathbb{E}_{\omega \sim \Pi_j(h)} [e^{-\Phi_j(h, \omega)} f(Y_j(h, \omega))],$$

*$U$  the uniform law. Let  $Q'$  be the law of the history built from the uniform law on  $R$  by drawing, at each block,  $\omega_j \sim \Pi_j(H_{<j})$  and appending  $Y_j$ . If the leak  $\mathcal{L} = Q'(\text{final history} \notin \mathcal{A})$  is at most  $1/2$ , then every majorant  $\nu$  of level  $\lambda$  satisfies*

$$\log \frac{1}{\mathbb{E}\nu} \leq \log \frac{Q_0}{|R|} + \log 2 + 2 \sum_j \mathbb{E}_{Q'} \Phi_j(H_{<j}, \omega_j).$$

*Proof.* Let  $g_j(h) = \mathbb{E}_U[\nu \mid H_{<j} = h]$ . By downward induction,  $g_j(h) \geq \mathbb{E}_{Q'}[\mathbf{1}_{\mathcal{A}} e^{-\sum_{i \geq j} \Phi_i} \mid H_{<j} = h]$ ; for  $j = J+1$  this is  $g_{J+1} \geq \mathbf{1}_{\mathcal{A}}$ . Given  $h$ ,  $f(y) = g_{j+1}(h, y)$  is  $\lambda$ -level and nonnegative, as in the proof of Theorem 7.4. Apply  $(S_w)$  and the induction hypothesis at  $(h, Y_j)$ ; under  $Q'$  the later blocks depend on  $\omega_j$  only through  $Y_j$ . Conclude with the Jensen step of Theorem 7.4, using  $\sum_j \Phi_j \geq 0$ .  $\square$

A step of Theorem 7.4 is an  $(S_w)$  with deterministic cost. The new instance is the *sequential step*: for a block  $V$  of primes with  $\log \ell > s$ ,  $\lambda$ -level functions are  $d$ -local with  $d = \lfloor \lambda/s \rfloor$ . Given  $h$ , the patterns are the classes with top prime in  $V$  whose requirements outside  $V$  are met by  $h$  (a class modulo  $\ell^v$  is a union of points of  $\mathbb{Z}/\ell^{E_\ell}$ ). Take for  $\Pi_j(h)$  the coupled process of Section 8 in increasing order with  $\varpi = U$  and caps  $\delta_\ell = \ell^{-1/2}$ ,  $Y_j = y$ , and for  $\Phi_j$  the cost of Theorem 8.5 with  $t = t_j(h) = d/(\mathbb{E}[m \mid h] + 4d)$ , fixed before the block is drawn. Then  $(S_w)$  is Theorem 8.5, and  $\Phi_j \geq 0$ .

**9.2. Inflation, moments and leak.** Fix  $B \geq 0$  and work with the base  $R_W$  of Lemma 7.2, caps  $\delta_\ell = \ell^{-1/2}$ , and  $\Gamma(m) = \prod_{p|m} \gamma'(p)$  with  $\gamma'(\ell) = (1 - \ell^{-1/2})^{-1}$  for  $\ell > W$  (Lemma 7.3(b)); assume  $W \geq 16$ , so that  $\delta_\ell \leq 1/4$ . The family  $\mathfrak{G}$  consists of  $\mathcal{R}(M)$ -classes with  $M \leq P(M)^{1+B}$ , of any shape, plus any  $\mathcal{R}(M)$ -classes of  $W$ -smooth modulus.

**Lemma 9.2** ([KA, Lemmas 4.2, 4.2', 4.3]; [PROVED]). *Let  $Q'$  be built with sequential steps in some blocks and singleton steps in the others.*

- (1) (inflation) Lemma 7.3(b) holds for  $Q'$ .
- (2) (first moment) There is  $K = K(W, B)$  such that for every  $s \geq \log W$  and every block  $V \subseteq \{\ell : s < \log \ell \leq 2s\}$ ,

$$\mathbb{E}_{Q'} m_V \leq \mathbb{E}_{Q'} \sum_{\ell \in V} p_\ell \leq \sum_{M: P(M) \in V} \frac{\tau(A_M^2) \Gamma(M)}{M} \leq K (2(1+B)s)^3,$$

the sum over the moduli of  $\mathfrak{G}$ ,  $A_M = (M+1)/4$ .

- (3) (second moment)  $\mathbb{E}_{Q'} p_\ell^2 \leq C(\epsilon, B) \ell^{-2+\epsilon}$  for every prime  $\ell > W$ .
- (4) (leak)  $\mathfrak{L} \leq \sum_\ell \mathbb{E}_{Q'} [p_\ell \mathbf{1}\{p_\ell > \delta_\ell\}] \ll_B W^{-1/4}$ , so  $\mathfrak{L} \leq 1/2$  for  $W \geq W_0(B)$ .

*Proof.* (1) Chain rule, with Lemma 8.1(2): given the past,  $y_\ell$  has density at most  $(1 - \delta_\ell)^{-1}$  with respect to  $U$ . (2) A class  $-4D \pmod{q\ell^v}$  with top  $\ell \in V$  adds at most  $\ell^{-v}$  to  $p_\ell$ , and only if  $n \equiv -4D \pmod{q}$  holds for the history before  $\ell$ , which has  $Q'$ -probability at most  $\Gamma(q)/q$  by (1). There are at most  $\tau(A_M^2)$  values of  $D$ . Write  $\Gamma(M) = \sum_{e|M} h(e)$  with  $h$  multiplicative on squarefree  $e$ ,  $h(p) \leq 2$  for  $p \leq W$ ,  $h(\ell) \leq 2\ell^{-1/2}$  for  $\ell > W$ . The weighted Shiu argument of Lemma 5.1 and Corollary 6.2 gives  $\sum_{M \leq x, M \equiv 3 \pmod{4}} \tau(A_M^2) \Gamma(M) \leq K_0(W) x \log^2(x+2)$ , with no windows and no  $B$ . Every  $M$  in the sum has  $M \leq P(M)^{1+B} \leq e^{2(1+B)s}$ ; drop the condition  $P(M) \in V$  and sum by parts. (3) is [TW, Lemma 4.0], the prime-power form of Lemma 7.5; it uses only (1). (4) A final history outside  $\mathcal{A}$  satisfies a class that is not  $W$ -smooth, by (R1). When its top prime  $\ell$  is processed its other requirements are met, so  $y_\ell$  lies in the completing set; this is impossible if  $\ell$  is light (Lemma 8.1(1)) and has conditional probability at most  $p_\ell$  if  $\ell$  is heavy. Markov and (3) give  $\sum_{\ell > W} \ell^{1/2} O_B(\ell^{-7/4})$ .  $\square$

The cubic growth  $s^3$  of the block mass, against the depth  $d \asymp \lambda/s$ , is what makes the sum over blocks below converge.

### 9.3. The cap.

**Theorem 9.3** (the 3/4 cap for fixed  $B$ ; [KA, Thm 4.5]; [PROVED]). *Fix  $B \geq 0$  and  $W = W_0(B)$ . There are  $\lambda_0(B)$  and  $C(B)$  such that for every family  $\mathfrak{G}$  of  $\mathcal{R}(M)$ -classes with  $M \leq P(M)^{1+B}$  (of any shape: dominant, gapped,  $\eta$ -twin, prime-power top, any number of prime factors at comparable scale), together with any  $\mathcal{R}(M)$ -classes of  $W$ -smooth modulus, every majorant  $\nu$  of the avoider set in  $R_W$  of level  $\lambda \geq \lambda_0(B)$ , charging all primes  $> W$ , satisfies*

$$\log \frac{1}{\mathbb{E}_\nu} \leq C(B) \lambda^{3/4}.$$

*Proof.* Put  $s_1 = \lambda^{1/4}$ , with  $\lambda_0$  such that  $s_1 > 2 \log W$ . The blocks, in increasing order of primes, are: singletons  $\{\ell\}$  for  $W < \ell \leq e^{s_1}$ ; sequential blocks  $V_i = \{\ell : 2^i s_1 < \log \ell \leq 2^{i+1} s_1\} \cap (1, e^{\lambda/2}]$  for  $0 \leq i \leq I$ ,  $I$  the largest index with  $2^I s_1 < \lambda/2$

(these cover  $(e^{s_1}, e^{\lambda/2}]$  exactly); one sequential block  $(e^{\lambda/2}, e^\lambda]$ ; and singletons above  $e^\lambda$ , of cost 0. Every class is decided at its top prime, and  $\mathfrak{L} \leq 1/2$  by Lemma 9.2(4), so Theorem 9.1 applies. The base costs  $O_B(1)$  (Lemma 7.2(2)). The singletons below  $e^{s_1}$  cost  $\leq \frac{8}{3}K'(1+B)^3\lambda^{3/4}$  in total [TW, Prop. 4.1], and the top block costs  $2\log(1+3e^{-\lambda/4})$  [TW, Lemma 4.2, Cor. 4.3]: there each term of  $\nu$  sees at most one prime. For  $V_i$  put  $s = 2^i s_1$  and  $d_i = \lfloor \lambda/s \rfloor \geq \lambda/(2s)$ . Corollary 8.6 with  $\bar{m} = \mathbb{E}[m \mid h]$  for each  $h$ , then Jensen over  $h$  (the map  $m \mapsto \log(C_0(m+4d)/d)$  is concave), give

$$\mathbb{E}_{Q'} \Phi_i \leq d_i \log \frac{C_0(\mathbb{E}m_{V_i} + 4d_i)}{d_i} + \frac{4}{3}d_i + \frac{1}{2}\log(22d_i + 22) + 3.$$

By Lemma 9.2(2),  $\mathbb{E}m_{V_i} \leq K_1 s^3$  with  $K_1 = 8(1+B)^3 K$ , so  $(\mathbb{E}m_{V_i} + 4d_i)/d_i \leq 2K_1 s^4/\lambda + 4 = 2K_1 \cdot 16^i + 4$ . Hence  $\mathbb{E}\Phi_i \leq 2^{-i}\lambda^{3/4}(c_1(B) + 4i\log 2) + O(\log \lambda)$ , and  $2\sum_i \mathbb{E}\Phi_i \leq 2\lambda^{3/4}\sum_i 2^{-i}(c_1(B) + 2.78i) + O(\log^2 \lambda) \ll_B \lambda^{3/4}$ .  $\square$

Theorem 9.3 has no family hypothesis beyond  $B$ : no gap, no window resolution, no bound on the number of prime factors, and no hypothesis on heavy coordinates. It removes the residual of Theorem 7.7, and it makes Conjecture 14.2, the hypothesis K2 and the extra  $\log \lambda$  of Proposition 14.3 unnecessary for the 3/4 question [KA, §4.3]. The constants  $W_0(B)$ ,  $\lambda_0(B)$  and  $C(B)$  are not tracked.

#### 9.4. The main theorem.

**Theorem 9.4** (the 3/4 cap for CRT-majorant architectures; [PROVED]; for Case-A classes [PROVED MOD] Elsholtz–Tao Prop. 1.4). *Fix  $A \geq 1$ . Let  $\mathfrak{F}$  be a family of forced classes such that either*

- (i) *for some fixed  $0 < C < 1$ , every modulus has a dominant prime  $P(M) \geq M^{1/(1+C)}$  (Case-B and Case-A classes; Corollary 6.2); or*
- (ii) *for some fixed  $B \geq 0$ ,  $\mathfrak{F}$  consists of  $\mathcal{R}(M)$ -classes with  $M \leq P(M)^{1+B}$ , of any shape, together with any  $\mathcal{R}(M)$ -classes of  $W_0(B)$ -smooth modulus (Theorem 9.3).*

*Then every CRT-majorant architecture (Definition 3.3) for  $\mathfrak{F}$ , with the admissible set of Corollary 6.2 in case (i) and the quadratic-residue base  $R_{W_0(B)}$  in case (ii), saves at most*

$$\log \frac{1}{\mathbb{E}\nu} \ll (\log N)^{3/4},$$

*with implied constant depending on  $A, C$  in case (i) and on  $A, B$  in case (ii). In particular no such architecture gives  $E(N) \leq N \exp\{-(\log N)^{3/4}\omega(N)\}$  with  $\omega(N) \rightarrow \infty$ , e.g.  $\omega = (\log \log N)^\epsilon$ .*

*Proof.* Corollary 3.5, applied with Corollary 6.2 for  $A+2$  in place of  $A$  in case (i), and with Theorem 9.3 at  $\lambda = (A+2)\log N \geq \lambda_0(B)$  in case (ii). In both cases the  $R$ -term is bounded.  $\square$

Case (ii) contains the gapped families of Theorem 7.6 and the window-resolved families of Theorem 7.7 (with an  $\eta$ -free constant), and the twin moduli of Lemma 13.1. For  $\mathcal{R}(M)$ -classes it also contains case (i) with  $B = C$ . The two cases are not combined: a family mixing  $(a, D)$ - or Case-A classes without a dominant prime with  $\mathcal{R}(M)$ -classes is not covered, because Lemma 2.6, and hence the base  $R_W$ , is proved for  $\mathcal{R}(M)$  only (Section 13). For the prime-slice families of Corollary 6.1 the

statement is the same with the selector term  $\log(P/\varphi(P))$  added; for the large sieve over slice systems the level hypothesis  $\lambda \leq A \log N$  is used directly (Remark 4.8).

## 10. THE CAMPAIGN SIEVES LIE IN THE CLASS

### 10.1. The three-quarter note.

**Lemma 10.1** ([ET, Lemma 4.1]; [PROVED]). *The atom family  $\mathcal{A}_X$  of [TQ, §2] is a prime-slice system with  $Q_0 = \text{lcm}(L_K, P_y)$ ,  $\mathcal{P}$  the primes in  $(X^{1/2}, X]$ , the selector  $R = \{(c, P_y) = 1\}$ ,  $|F_\ell(c)| \leq \ell^{1/3}$ , and  $\sum_\ell p_\ell(c) \ell^{-\alpha} \leq C_u t^3 X^{-\alpha/2}$  for all  $c$  (the upper half of [TQ, Cor. 4.3], “uniform fibre masses”), where  $t = \log X$ . Its majorant  $\nu_X = S_y Q_r(H_X)$  is  $\geq 0$  on  $\mathbb{Z}$ , equals 1 on the whole avoider set  $\mathcal{A}$  (there  $S_y = 1$ ,  $H_X = 0$  and  $Q_r(0) = 1$ ; [ETr, item 3.1]; [TQ, Thm 8.2] itself states only  $\nu_X \geq 1$  on exceptional primes), and has level  $\lambda = rt$ . Hence every majorant on this family has*

$$\log \frac{1}{\mathbb{E}\nu} \leq \log \log y + \min \left\{ Ct^3, C \frac{\lambda}{t} \left( 1 + \log^+ \frac{Ct^4}{\lambda} \right) \right\} + O(\log^2 \lambda).$$

*Proof.* The structure is [TQ, Lemma 2.2] (deduplication and conditional independence): conditional on  $c \bmod L_K$  the coordinates  $n \bmod \ell$  are independent and uniform, and an atom  $(k, \ell, u, v)$  forbids  $-uv^{-1} \bmod \ell$  in the fibres with  $u + cv \equiv 0 \pmod{k}$ ; at most  $z_j^2 \leq \ell^{1/3}$  atoms live at each  $\ell$ . The majorant is a product of the selector  $S_y$  and  $Q_r(H) = \binom{H-1}{r}$ , both nonnegative, and  $Q_r(0) = 1$ . A term of  $Q_r(H_X)$  involves at most  $r$  atoms, hence at most  $r$  slice primes  $\leq X$ , so its level is at most  $rt$ . The bound is Theorem 4.5 with  $\alpha$  optimised and the selector term  $\log(P_y/\varphi(P_y)) = \log \log y + O(1)$ .  $\square$

With  $\lambda \leq A \log N$  the maximum over  $t$  is  $\asymp (\log N)^{3/4}$ , attained only for  $t \asymp (\log N)^{1/4}$ . What forces  $\lambda \lesssim \log N$  is not the note’s modulus bound, which it calls “not necessary”, but its coefficient budget  $T_{\text{abs}} \leq N^{1/2}$  [TQ, proof of Thm 8.2]: the slice primes are  $\leq X$ , so  $\Lambda_0 = t \leq \log N$ , and Lemma 3.4 converts any majorant on this family with coefficient sum  $< N$  into one of level  $\leq t + \log N + s$  with at most twice the mean. So the note satisfies (A1)–(A3), and *its exponent 3/4 is sharp for its own architecture* [PROVED]. This was the review’s one substantive correction [ETr, item 3, SC1].

**Lemma 10.2** (Bonferroni depth; [ET, Lemma 4.2]; [PROVED]). *For even  $r$  and any integer  $H \geq 0$ ,  $\mathbb{E}Q_r(H) \geq \mathbb{P}(H \geq r+1)$ . If  $H$  is a sum of independent Bernoulli variables with mean  $\mu$  and  $r+1 \leq \mu - 2\sqrt{\mu}$ , then  $\mathbb{E}Q_r(H) \geq 3/4$ .*

*Proof.*  $Q_r(h) = \binom{h-1}{r} \geq 1$  for  $h \geq r+1$ , and Chebyshev with  $\text{Var } H \leq \mu$ .  $\square$

So the depth  $r \gg t^3$  in [TQ] is forced. The binding constraint in the 3/4 proof is the pair (cubic supply  $\mu_c \leq C_u t^3$ , budget  $\log T_{\text{abs}} \leq \log N$ ). Bonferroni depth, the Bombieri–Vinogradov level, the selector and the pruning cannot improve the exponent: the cap of Theorem 4.5 holds for every majorant, and Lemmas 5.1–5.3 bound the *entire* forced-class supply independently of any level of distribution, so Elliott–Halberstam cannot raise the profile either [ET, Lemma 4.3 and remark] [PROVED]. That Selberg or other majorants *attain* the same order up to constants is only [EVIDENCE]: in an exchangeable Poisson model the numerically computed linear-programming optimum agrees with the Selberg value to within 0.01 in  $-\log$  in the tested cases [ET, §2.5].

**10.2. The two-thirds loglog note.** The note [LL] is not a majorant sieve but Montgomery's large sieve over progressions modulo  $L_K$ , with multipliers  $K = \{k \leq \delta \log N\}$ . It is covered through Remark 4.8.

**Corollary 10.3** (restricted multiplier sets; [ET, Cor. 3.5]; [PROVED] given the notes' Brun–Titchmarsh upper bounds). *Suppose a family at scale  $X = e^t$  uses slice primes in  $(X^{1/2}, X]$  and multipliers  $k \in K$ , with  $\sum_{\ell} p_{\ell}(c) \leq A_0 t^2 h(K)$  for all  $c$ , where  $h(K) = \sum_{k \in K} \varphi(k)/k^2$ . Then*

$$\text{saving} \leq \min \left\{ Ct^2 h, \frac{38\lambda}{t} \left( 1 + \log^+ \frac{Ct^3 h}{\lambda} \right) \right\} + O(\log^2 \lambda),$$

and the supremum over  $t$  is  $\asymp \lambda^{2/3} h^{1/3}$ . For  $K = \{k \leq \delta \log N\}$ ,  $h \asymp \log \log N$ , giving  $(\log N)^{2/3} (\log \log N)^{1/3}$ ; the same holds for any  $K$  with  $\text{lcm}(K) \leq N$ , since then  $h(K) \leq \prod_{p|\text{lcm}} (1 + 1/p) \ll \log \log N$ .

*Proof.* Insert  $\sum_{\ell} \bar{p}_{\ell} \ell^{-\alpha} \leq A_0 t^2 h e^{-\alpha t/2}$  into (4). With  $t_0 = (\lambda/h)^{1/3}$ , for  $t \leq t_0$  the first entry is  $\leq C \lambda^{2/3} h^{1/3}$ ; for  $t = ut_0 \geq t_0$  the second is  $\leq C \lambda^{2/3} h^{1/3} (1 + 3 \log u)/u$ .  $\square$

The note's large-sieve level is  $Q^2 \leq N/L_K$ , so  $\lambda \leq \log N$ , and its admissible set is the reduced residues modulo  $L_K$ , with  $R$ -term  $\log(L_K/\varphi(L_K)) \ll \log \log \log N$ . By Remark 4.8 and Corollary 10.3 its saving is at most  $C(\log N)^{2/3} (\log \log N)^{1/3}$ , i.e. its bound is never smaller than  $N \exp\{-C(\log N)^{2/3} (\log \log N)^{1/3}\}$ : the 2/3-loglog note is sharp for its architecture [ET, Lemma 4.4], [PROVED]. Vaughan's argument and that of Pomerance–Weingartner [PW] are the case  $K = \{1\}$ , giving  $(\log N)^{2/3}$ . For the 3/4 note,  $h \asymp \kappa t$  and the same computation gives  $\lambda^{3/4}$ .

## 11. SELBERG-TYPE MAJORANTS

**11.1. A sieve limit with no structural hypothesis.** Let  $\Omega = \prod_i \Omega_i$  be a finite product probability space (CRT coordinates), with costs  $s_i \geq 0$ :  $s_i = \log \ell$  for a slice prime and  $s_i = 0$  for free coordinates. Let  $\mathcal{A} \subseteq \Omega$  be arbitrary. Let  $V_{\kappa}$  be the span of functions of  $\omega_T$  over sets  $T$  with  $\sum_{i \in T} s_i \leq \kappa$ ; it is the sum of the Efron–Stein components  $H_T$  over these  $T$ .

**Theorem 11.1** (Selberg-type sieve limit; [ET, Thm 5.5]; [PROVED]). *Let  $g \in V_{\lambda/2}$  with  $g \geq 1$  on  $\mathcal{A}$ , so that  $\nu = g^2$  is a majorant of level  $\lambda$ . Then for every  $\alpha > 0$ ,*

$$\mathbb{E} g^2 \geq e^{-\alpha \lambda/2} \frac{\mathbb{P}(\mathcal{A})^2}{\mathbb{P}(\omega \in \mathcal{A}, \omega' \in \mathcal{A})},$$

where  $\omega'$  is the  $\rho$ -correlated copy: independently for each  $i$ ,  $\omega'_i = \omega_i$  with probability  $\rho_i = e^{-\alpha s_i}$  and otherwise a fresh sample. Equivalently,  $\log(1/\mathbb{E} \nu) \leq \alpha \lambda/2 + \Xi_{\mathcal{A}}(\alpha)$  with the noise-stability excess  $\Xi_{\mathcal{A}}(\alpha) = \log[\mathbb{P}(\mathcal{A} \cap \mathcal{A}'_{\alpha})/\mathbb{P}(\mathcal{A})^2] \in [0, \log(1/\mathbb{P}(\mathcal{A}))]$ .

*Proof.*  $\mathbb{E}[g \mathbf{1}_{\mathcal{A}}] \geq \mathbb{P}(\mathcal{A})$  and, since  $g \in V = V_{\lambda/2}$ ,  $\mathbb{E}[g \mathbf{1}_{\mathcal{A}}] = \langle g, \Pi_V \mathbf{1}_{\mathcal{A}} \rangle \leq \|g\| \|\Pi_V \mathbf{1}_{\mathcal{A}}\|$ . Now

$$\begin{aligned} \|\Pi_V \mathbf{1}_{\mathcal{A}}\|^2 &= \sum_{c(T) \leq \lambda/2} \|(\mathbf{1}_{\mathcal{A}})_T\|^2 \leq e^{\alpha \lambda/2} \sum_T e^{-\alpha c(T)} \|(\mathbf{1}_{\mathcal{A}})_T\|^2 \\ &= e^{\alpha \lambda/2} \langle \mathbf{1}_{\mathcal{A}}, T_{\rho} \mathbf{1}_{\mathcal{A}} \rangle = e^{\alpha \lambda/2} \mathbb{P}(\mathcal{A} \cap \mathcal{A}'), \end{aligned}$$

where  $c(T) = \sum_{i \in T} s_i$  and the noise operator  $T_{\rho}$  acts on  $H_T$  by  $\prod_{i \in T} \rho_i$ .  $\square$



This covers Selberg's  $\Lambda^2$  sieve. For a prime-slice system the excess factorises fibrewise,  $\Xi_{\mathcal{A}_c}(\alpha) = \sum_{\ell} \log(1 + \rho_{\ell} p_{\ell}(c)/(1 - p_{\ell}(c))) \leq \frac{4}{3} \sum_{\ell} p_{\ell}(c) \ell^{-\alpha}$  when  $p_{\ell}(c) \leq 1/4$ , so Theorem 11.1 reproduces the Rankin functional [ET, Cor. 5.6]. Its point is that balanced, twin and shared-prime moduli are allowed: for  $\Lambda^2$ -majorants, beating  $3/4$  with balanced moduli requires the following hypothesis to fail.

**Hypothesis 11.2** ( $H_{\text{MS}}^{\text{Sel}}$ ; [ET, §5.7]; [OPEN]). For the complete Case-B and Case-A forced-class system with moduli  $\leq e^{\lambda}$ , with primes  $\leq \lambda^4$  given  $\rho = 1$ ,  $\Xi_{\mathcal{A}}(\alpha) \ll \alpha^{-3}(\log \lambda)^{O(1)}$  uniformly for  $\lambda^{-1} \leq \alpha \leq 1$ .

Under Hypothesis 11.2, Theorem 11.1 caps Selberg-type majorants at  $\lambda^{3/4}(\log \lambda)^{O(1)}$  for all moduli [CONDITIONAL]. A fibrewise version alone would not suffice:  $\mathcal{A} = \{c = c_0\} \times \Omega'$  has  $\Xi = 0$  in every nonempty fibre, yet  $g = \mathbf{1}_{c=c_0}$  saves  $\log Q_0$ . The remedy is to tilt the fibre law.

**Lemma 11.3** (fibre tilting; [TW, Lemma 6.6], [TW2, Lemma 2.1]; [PROVED]). *In the setting of Theorem 11.1, let  $F$  be a set of coordinates whose residues  $c$  (the fibre) range over  $\mathbb{Z}/Q_F$ ; put  $\mathcal{A}_c = \mathcal{A} \cap \{\text{fibre} = c\}$  and  $\Xi_c(\alpha) = \log[\mathbb{P}(\mathcal{A}_c \cap \mathcal{A}'_c \mid c)/\mathbb{P}(\mathcal{A}_c \mid c)^2]$ , where in  $\mathcal{A}'_c$  the coordinates outside  $F$  are  $\rho$ -correlated and the fibre is shared. For any probability law  $P$  on fibres with  $\mathcal{A}_c \neq \emptyset$  on its support, every  $g \in V_{\lambda/2}$  with  $g \geq 1$  on  $\mathcal{A}$  satisfies*

$$\log \frac{1}{\mathbb{E}g^2} \leq \frac{\alpha\lambda}{2} + \log \left\| \frac{dP}{dU} \right\|_{\infty} + \mathbb{E}_P \Xi_c(\alpha).$$

For  $P$  uniform on a set  $R$  the middle term is  $\log(Q_F/|R|)$ .

*Proof for uniform  $P$  on  $R$ .* Let  $\tilde{\sigma} = \sum_{c \in R} \pi_c U(\cdot \mid c, \mathcal{A})$  and  $h = d\tilde{\sigma}/dU$ . Then  $\mathbb{E}_U[gh] \geq 1$ , so  $\mathbb{E}g^2 \geq 1/\|\Pi_{V_{\lambda/2}} h\|^2$ , and  $\|\Pi_V h\|^2 \leq e^{\alpha\lambda/2} \langle h, T_{\rho} h \rangle$ . Since  $\langle h, T_{\rho} h \rangle$  is nondecreasing in each  $\rho_i$ , put  $\rho_i = 1$  on  $F$ ; then  $\langle h, T_{\rho} h \rangle = Q_F \sum_{c \in R} \pi_c^2 e^{\Xi_c}$ . Choosing  $\pi_c \propto e^{-\Xi_c}$  and Jensen give  $\leq (Q_F/|R|) e^{\text{avg}_R \Xi_c}$ . The general case is [TW2, Lemma 2.1].  $\square$

**11.2. Two large primes, twins included.** *Setting* [TW2, §3]. Fix  $B \geq 0$ ,  $A_0 \geq 1$ ; let  $X$  be large,  $L = \log X$ ,  $W_1 = L^{1/2}$ ,  $w_2 = L^8$  and  $\alpha = L^{-1/4}$ . The family is any set of classes  $-4D \pmod{M}$  of  $\mathcal{R}(M)$  with  $M \leq X$ ,  $M \leq P(M)^{1+B}$ , and at most two distinct prime factors of  $M$  above  $w_2$ . Twin moduli  $k\ell_1\ell_2$  ( $\ell_i > w_2$ ,  $k$   $w_2$ -smooth), dominant moduli and prime powers are included. All primes are charged, and the level is  $\lambda \leq A_0 L$ . The fibre is  $c = n \pmod{w_2}$ -smooth part, with a law  $P$  built from the QR base at primes  $\leq W_1$ , conditioning on avoidance of the small classes, and a good-fibre event; then  $\log \|dP/dU\|_{\infty} \leq 4L^{1/2}$  [TW2, Lemmas 3.1–3.4]. In a fibre, a class  $k\ell^v$  is unary at  $\ell$  and a class  $k\ell^v m^u$  is a binary edge.

The probabilistic core is a binary noise-stability bound. Take independent coordinates  $y_{\ell}$  with laws  $\nu_{\ell}$  and a finite set of edges  $e = \{(\ell, a), (m, c)\}$ ,  $\ell \neq m$ , hit when  $y_{\ell} = a$  and  $y_m = c$ . Put  $\pi_e = \nu_{\ell}(a)\nu_m(c)$ ,  $\deg(\ell, a) = \sum \nu_m(c)$  over the edges at  $(\ell, a)$ ,  $w_{\ell} = \sum_{e \ni \ell} \pi_e$  and  $q_{\ell} = \sum_a \nu_{\ell}(a) \deg(\ell, a)^2$ , and assume  $w_{\ell} \leq \delta \leq 1/16$  for all  $\ell$ . Let  $Z_1$  be the probability that no edge is hit, and  $Z_2(\tilde{\rho})$  the probability that no edge is hit in either copy of the coupled pair in which, independently for each  $\ell$ ,  $y'_{\ell} = y_{\ell}$  with probability  $\tilde{\rho}_{\ell}$  and is otherwise an independent copy.

**Theorem 11.4** (binary noise stability; [TW2, Thm 1.4]; [PROVED]). *For every  $\tilde{\rho} \in [0, 1]^{\text{index set}}$ ,*

$$\log \frac{Z_2(\tilde{\rho})}{Z_1^2} \leq (1 + 25\delta) \left[ \sum_{e=\{\ell, m\}} \tilde{\rho}_\ell \tilde{\rho}_m \pi_e + \sum_j \tilde{\rho}_j q_j \right].$$

The proof [TW2, §1] differentiates  $\log Z_2$  along  $\kappa = t\tilde{\rho}$ , uses an identity for  $\partial_{\kappa_j} \log Z_2$  under the unconditioned law of the other coordinates, and bounds the result with a conditional form of the Lovász local lemma; no cluster expansion is needed. An exact check on 1200 small systems found no violation [EVIDENCE].

For a fibre  $c$ , a prime  $j > w_2$  and a residue  $a$ , let  $\deg_c(j, a)$  be the partner mass of the active binary classes through  $(j, a)$ , and  $S_j(c) = \sum_a \nu_j(a) \min(\deg_c(j, a), 1)^2$ .

**Theorem 11.5** (two-prime  $\Lambda^2$  cap, reduced; [TW2, Thm 5.1]; [PROVED]). *In this setting, for  $L$  large, every  $g \in V_{\lambda/2}$  with  $g \geq 1$  on the avoiders of the family satisfies*

$$\log \frac{1}{\mathbb{E}g^2} \leq \frac{A_0}{2} L^{3/4} + C_B L^{3/4} (\log L)^C + 11 \mathbb{E}_P \sum_{j>w_2} \rho_j S_j(c), \quad \rho_j = j^{-\alpha},$$

with an absolute exponent  $C$ .

The proof combines Lemma 11.3 with the fibre law  $P$ , Theorem 11.4 in each good fibre after a free quarantine of hub residues [TW2, Lemma 2.2], and a profile bound  $\ll \alpha^{-3}(\log L)^{O(1)}$  for the unary and diagonal binary terms [TW2, Lemma 4.1].

The off-diagonal term was left open in [TW2]. It is now bounded unconditionally [TW3]; the review [TW3r] found the argument sound after repairs.

**Theorem 11.6** (the off-diagonal term  $(H_O)$ ; [TW3, Thm 4.1]; [PROVED]). *In this setting,  $\mathbb{E}_P \sum_{j>w_2} \rho_j S_j(c) \ll_B \alpha^{-3}(\log L)^{O(1)}$ .*

*Proof sketch.* The prime-power classes are split off as in [TW2, Lemma 5.4], at cost  $o(1)$ . For the classes with moduli  $kjm$  ( $j \neq m$  primes  $> w_2$ ,  $k$   $w_2$ -smooth), call a class through  $j$  *small* if  $m \leq (kj)^{C_0}$  and *large* otherwise, for a fixed  $C_0 \geq 6$ , and write  $\deg_c(j, a) = x(a) + z(a)$  accordingly. Since  $\min(x + z, 1) \leq \min(x, 1) + z$  and  $\min(x, 1)^2 \leq x$ ,

$$S_j \leq 2w_j^{\text{sm}}(c) + 2 \sum_a \nu_j(a) z(a)^2,$$

where  $w_j^{\text{sm}}$  is the binary mass at  $j$  of the small active classes. *Small part* [TW3, Lemma 2.1]: by Shiu's theorem along the top prime, the divisor mass of a small class is  $(\log kj)^3$  rather than  $L^3$ , and  $\sum_j \rho_j (\log j)^3 / j \ll \alpha^{-3}$ ; so its first moment is  $\ll \alpha^{-3}(\log L)^{O(1)}$ . *Large part* [TW3, Lemmas 3.1–3.4]: parametrise  $D \mid A^2$  by  $A = uv$ ,  $(u, v) = 1$ ,  $D = u^2 t$ . The largest of  $u, v, t$  is at least  $A^{1/3}$ , and  $m > (kj)^{C_0}$  makes the range of the prime  $m$  long enough for Brun–Titchmarsh in a class modulo four times the product of the two shorter variables. The residue modulo  $j$  depends only on those two variables, and their second moments give  $\sum_{a \bmod j} V_{j,k}(a)^2 \ll (\log L)^6$  uniformly in  $j, k$ . Activity and the  $k$ -sum are handled as in [TW2, Lemma 5.4]. The large part contributes  $\ll (\log L)^{O(1)}$ .  $\square$

**Theorem 11.7** (two-prime  $\Lambda^2$  cap, twins included; [TW2, Cor. 5.2], [TW3, Cor. 4.2]; [PROVED]). *Fix  $B \geq 0$  and  $A_0 \geq 1$ . Let the family consist of classes of  $\mathcal{R}(M)$  with  $M \leq X$ ,  $M \leq P(M)^{1+B}$ , and at most two distinct prime factors of  $M$  above*

$(\log X)^8$  (twin moduli  $k\ell_1\ell_2$ , dominant moduli and prime powers included). Then every  $g \in V_{\lambda/2}$ ,  $\lambda \leq A_0 \log X$ , with  $g \geq 1$  on the avoiders of the family satisfies

$$\log \frac{1}{\mathbb{E}g^2} \ll_{A_0, B} L^{3/4} (\log L)^{O(1)}, \quad L = \log X.$$

*Proof.* Theorems 11.5 and 11.6.  $\square$

*Scope.* This is a statement about Selberg-type majorants  $g^2$  only. It needs at most two prime factors above  $(\log X)^8$ ; the extension to any fixed number is Theorem 11.11 below. The restriction  $M \leq P(M)^{1+B}$  with  $B$  fixed excludes, for fixed  $B$ , twins  $k\ell_1\ell_2$  with a  $w_2$ -smooth cofactor  $k > M^{(B-1)/(B+1)}$ ; for a fixed number of large primes it is removed by Theorem 11.12. The bound is stated in  $L = \log X$ , with moduli  $\leq X$  and level  $\lambda \leq A_0 L$ . The level hypothesis  $\lambda \leq A_0 L$  is assumed directly: the coarsening of Lemma 3.4 does not preserve the square form  $g^2$ , so the reduction from a coefficient budget to a level (Corollary 3.5) is not available here as stated (observation of this note, not from [TW2, TW3]).

*Remark 11.8* (relation to the main theorem). A  $\Lambda^2$ -majorant  $g^2$  with  $g \in V_{\lambda/2}$  and  $g \geq 1$  on the avoiders of a family is a nonnegative majorant of level  $\lambda$ , and it is  $\geq 1$  on the avoiders that lie in  $R_W$ . So for families with  $M \leq P(M)^{1+B}$ ,  $B$  fixed, Theorem 9.3 applies to  $g^2$  directly and gives  $\log(1/\mathbb{E}g^2) \ll_B \lambda^{3/4}$  for  $\lambda \geq \lambda_0(B)$ , with any number of large prime factors and no power of  $\log L$ . Under the  $B$ -hypothesis, Theorems 11.7 and 11.11 are therefore superseded as caps [TW4, §11]. What the  $\Lambda^2$  route still gives beyond Theorem 9.3 is the removal of  $B$  for a fixed number of large primes (Theorem 11.12), and the noise-stability tools themselves.

*Remark 11.9* (what was bypassed). The route of [TW2] split the off-diagonal term into a same-label part  $(H_O^-)$  (proved there, [TW2, Lemma 5.4]) and a cross-label part  $(H_O^\neq)$ , both as *un-quarantined* pair sums, with a sufficient form  $H_{\text{div}}$  [TW2, §5.5]. Theorem 11.6 does not prove  $(H_O^\neq)$  or  $H_{\text{div}}$ : small partners are paid for through  $\min(x, 1)^2 \leq x$ , which says nothing about pair sums. Both remain [OPEN] as stated, but are no longer needed. The exponent budget  $\alpha^{-3}$  is used only by the small part, so this route does not lower the exponent  $3/4$ .

**11.3. Any fixed number of large primes, and removing  $B$ .** The results of this subsection are from [TW4]; the review [TW4r] (two rounds) found them sound after nits and one minor repair (G1), all applied. Their new arithmetic input is an upper-bound sieve for the *whole* partner of a modulus, from the arithmetic large sieve, in place of Brun–Titchmarsh for a prime partner.

**Lemma 11.10** (rough-partner Brun–Titchmarsh; [TW4, Lemma 2.3]; [PROVED]). *Let  $s \geq 1$ ,  $w \geq 2$ ,  $q \geq 1$ ,  $(b, q) = 1$  and  $x \geq q2^{s+1}$ . Let  $\mathcal{R}_s(w)$  be the set of integers all of whose prime factors exceed  $w$  and with  $\Omega(R) \leq s$ . Put  $Y = x/q$ ,  $Z = Y^{1/(s+1)}$  and  $H = \sum_{w < p \leq Z} 1/(p-1)$  ( $H = 0$  if  $Z \leq w$ ). Then*

$$\sum_{\substack{R \in \mathcal{R}_s(w), \ x < R \leq 2x \\ R \equiv b \pmod{q}}} \frac{1}{R} \leq \frac{3(s+1)(1+H+\dots+H^{s-1})}{\varphi(q) \log Y}.$$

*Proof.* Write  $R = dn$ ,  $d$  the part of  $R$  at primes  $\leq Z$ . If  $d = R$  then  $R \leq Z^s < x$ ; so  $\Omega(d) \leq s-1$ ,  $d \leq Z^{s-1}$ ,  $(d, q) = 1$ , and  $n > 1$  has all prime factors  $> Z$ , with  $n \equiv b\bar{d} \pmod{q}$  and  $n \in (x/d, 2x/d]$ . Fix  $d$  and write  $n = n_0 + qk$ . The

admissible  $k$  lie in an interval of at most  $x/(dq) + 1$  consecutive integers and avoid one class modulo each prime  $p \leq Z$ ,  $p \nmid q$ . The arithmetic large sieve [MV] (in the form  $|\mathcal{N}| \leq (N + Q^2)/J$  with  $Q = Z$ , and  $J \geq (\varphi(q)/q) \log Z$  by [vLR], give  $\#\{n\} \leq (x/(dq) + 1 + Z^2)/((\varphi(q)/q) \log Z) \leq 3x/(d\varphi(q) \log Z)$ , since  $x/(dq) \geq Z^2$  and  $x/(dq) \geq 1$ . Each  $R$  has  $1/R < 1/x$ . Sum over  $d$ :  $\sum_d 1/d \leq \sum_{i < s} H^i$ , and  $\log Z = \log Y/(s + 1)$ .  $\square$

For  $s = 1$  this is Brun–Titchmarsh up to the constant. Neither the primality of the partner nor its splitting is used. In the proof of Theorem 11.11 below it takes the place of the Brun–Titchmarsh step used in Theorem 11.6, and it closes the ternary residual left open in [TW3, §6.3]: there the partner  $R = \ell_a \ell_b$  with  $\ell_b < w_2 q$  is itself rough to level  $(R/q)/w_2$ , so only an *upper* bound for rough integers in one progression of length  $R/q$  is needed, not an asymptotic of Bombieri–Friedlander–Iwaniec type [TW4, §2].

**Theorem 11.11** ( $r$ -prime  $\Lambda^2$  cap; [TW4, Thm 7.1]; [PROVED]). *Fix  $r \geq 2$ ,  $B \geq 0$  and  $A_0 \geq 1$ . Let the family consist of classes of  $\mathcal{R}(M)$  with  $M \leq X$ ,  $M \leq P(M)^{1+B}$ , and at most  $r$  distinct prime factors of  $M$  above  $w_2 = (\log X)^8$ . Then every  $g \in V_{\lambda/2}$ ,  $\lambda \leq A_0 \log X$ , with  $g \geq 1$  on the avoiders of the family satisfies*

$$\log \frac{1}{\mathbb{E}g^2} \ll_{A_0, B, r} L^{3/4} (\log L)^{3r+O(1)}.$$

*Proof sketch.* The noise-stability input is the any-arity bound with codegree terms and free codegree quarantine of [TW3, Lemmas 6.1, 6.2]; it reduces the saving to star sums [TW4, Prop. 3.1]. The fibre law of [TW2, §3] extends with threshold  $\delta_r$  [TW4, Lemma 4.1]. Whole-event and unary terms are bounded by Shiu along the top prime ( $\ll \alpha^{-3}(\log L)^{r+O(1)}$ ), prime-power classes cost  $o(1)$ , small partners are paid through  $\min(x, 1)^2 \leq x$ , and large partners through a largest-variable reduction with Lemma 11.10 and box counting modulo squarefree  $Q$ ,  $\ll (\log L)^{3r+O(1)}$  [TW4, §§5–6].  $\square$

The  $r$ -dependence is explicit [TW4, Prop. 9.1] [PROVED]: every implied constant is  $\leq (C_B \log L)^{Cr}$ , so for  $r \leq \epsilon \log L / \log \log L$ ,  $\epsilon \leq \epsilon_0$ , the cap is  $\ll_{A_0, B} L^{3/4+C\epsilon}$ . At the other end, classes with at least  $330 \log L$  prime factors above  $w_2$ , with no  $B$ -hypothesis, can be added to any such family at cost  $o(1)$ , provided the fibre law also conditions on a low-mass event [TW4, Lemma 9.2, Cor. 9.3] [PROVED].

**Theorem 11.12** (no  $B$ -hypothesis for fixed  $r$ ; [TW4, Thm 10.4]; [PROVED]). *Fix  $r \geq 2$  and  $A_0 \geq 1$ . For families of  $\mathcal{R}(M)$ -classes with  $M \leq X$  and at most  $r$  distinct prime factors above  $(\log X)^8$ , with no condition relating  $M$  to  $P(M)$ , every  $g \in V_{\lambda/2}$ ,  $\lambda \leq A_0 \log X$ , with  $g \geq 1$  on the avoiders satisfies*

$$\log \frac{1}{\mathbb{E}g^2} \ll_{A_0, r} L^{3/4} (\log L)^{O_r(1)}.$$

*The same holds if the family also contains arbitrary  $\mathcal{R}(M)$ -classes with at least  $330 \log L$  prime factors above  $(\log X)^8$ .*

*Proof sketch.* Write  $M = kM_L$  with  $k$  the  $w_2$ -smooth part. The classes split into *smooth-dominated* ( $k > M_L^2$ ), *high-power* (not smooth-dominated,  $\ell^{12} \mid M$  for some  $\ell > w_2$ ), and the rest, which satisfy  $M \leq P(M)^{33r}$  and are handled by Theorem 11.11 with  $B = 33r - 1$ . For the smooth-dominated classes, Cauchy–Schwarz, Rankin’s trick on the smooth  $k$  and Shiu’s theorem along  $k$  (the modulus  $M_L$  is now short)

give a polylogarithmic *unweighted* first moment [TW4, Lemmas 10.1, 10.2]; the high-power classes have first moment  $\ll L^{-3}$ . The fibre law survives without  $B$  [TW4, Lemma 10.3].  $\square$

The middle range  $\epsilon \log L / \log \log L < r < 330 \log L$  is not covered, and it carries most of the  $\alpha^{-3}$ -weighted mass: there the method needs a hub count and an off-diagonal pair-sum bound that are efficient per prime [TW4, §§9.3, 12] [OPEN].

## 12. BEYOND CRT EVALUATION: ROUNDING, PRIMES AND MOMENTS

The caps above concern bounds of the shape  $N\mathbb{E}\nu + \sum_i |a_i|$  for majorants  $\geq 0$  on  $\mathbb{Z}$  and  $\geq 1$  on the whole avoider set. This section treats three natural ways out: signed rounding, majorants that are positive only at primes, and moment methods. Everything here is from [NC], for the *finite prime-slice* families of Corollary 6.1 only (not the sequential or balanced settings of Sections 7–9); the review [NCr] (two rounds) found it sound after repairs, which narrowed the scope statements reproduced below, and led to the withdrawal of one conjecture.

**12.1. Per-frequency rounding.** For a majorant  $\nu$  with common period  $Q_{\text{tot}}$  put  $\hat{\nu}(\theta) = \mathbb{E}_n \nu(n) e(-n\theta)$  for  $\theta \in Q_{\text{tot}}^{-1} \mathbb{Z} / \mathbb{Z}$ , so  $\hat{\nu}(0) = \mathbb{E}\nu$  and, exactly,

$$\sum_{n \leq N} \nu(n) - N\mathbb{E}\nu = \sum_{\theta \neq 0} \hat{\nu}(\theta) S_N(\theta), \quad S_N(\theta) = \sum_{n \leq N} e(n\theta). \quad (9)$$

A *per-frequency rounding bound* is  $R_w(\nu) = \sum_{\theta \neq 0} |\hat{\nu}(\theta)| w(\theta)$  with a weight  $w \geq 1$ . Every class indicator has  $\sum_{\theta} |\hat{\mathbf{1}}(\theta)| = 1$ , so  $R_1(\nu) \leq \sum_i |a_i|$ , and  $R_1$  can be much smaller. The weight  $w(\theta) = \min(N, 1/(2\|\theta\|))$  makes  $R_w$  a valid bound for (9), and complete Gauss, Kloosterman, Ramanujan or divisor-exponential sums over the classes sharing one frequency are already inside  $|\hat{\nu}(\theta)|$ .

For independent  $x_i \sim \text{Bern}(p_i)$  write  $y_i = x_i - p_i$ ; every  $f$  on  $\{0, 1\}^I$  has a unique biased Walsh expansion  $f = \sum_S d_S y^S$ ,  $d_{\emptyset} = \mathbb{E}f$ .

**Proposition 12.1** (Boolean sieve limit at arbitrary level; [NC, Prop. 2.1]; [PROVED]). *In the setting of Proposition 4.4, assume  $p_i \leq 1/4$  for all  $i$ , and let  $f \geq 0$  with  $f(0) \geq 1$  be of any level. Put  $r_0 = \sum_{s(S) > \lambda} |d_S| \prod_{i \in S} p_i$  and  $r_1 = \sum_{s(S) > \lambda} |d_S| \prod_{i \in S} 2p_i(1 - p_i)$ ,  $s(S) = \sum_{i \in S} s_i$ . Then  $\mathbb{E}f \geq (1 - r_0)e^{-\Phi} - 2r_1$ , with  $\Phi$  the right side of (3).*

*Proof sketch.* Split  $f = f_{\text{lo}} + f_{\text{hi}}$  at level  $\lambda$ . Then  $\mathbb{E}f = \mathbb{E}f_{\text{lo}}$ ,  $f_{\text{lo}}(0) \geq 1 - r_0$ ,  $\mathbb{E}|f_{\text{hi}}| \leq r_1$ , and  $f_{\text{lo}} \geq -h$  with  $h = \mathbb{E}[|f_{\text{hi}}| \mid x_{\{s_i \leq \lambda\}}]$ . Run the proof of Proposition 4.4 on  $f_{\text{lo}}$ , carrying  $h$ : at the interpolation step write  $LQ \leq |L|(Q + H) + |L|H$  with  $Q + H \geq 0$ .  $\square$

**Lemma 12.2** (Walsh coefficients versus Fourier mass; [NC, Lemma 2.2]; [PROVED]). *Take a prime-slice system with  $\mathcal{P}$  finite and a majorant  $\nu$  (any moduli). For  $c \in R$  let  $\nu_c = \sum_S d_S(c) y^S$  be the conditional expectation of  $\nu$  given  $n \equiv c \pmod{Q_0}$  and the hit indicators. For  $S \neq \emptyset$  let  $\Theta_S$  be the frequencies  $a/Q_0 + \sum_{\ell \in S} h_{\ell}/\ell$  with every  $h_{\ell} \not\equiv 0 \pmod{\ell}$ , and  $A_S(\nu) = \sum_{\theta \in \Theta_S} |\hat{\nu}(\theta)|$ . Then  $|d_S(c)| \prod_{\ell \in S} p_{\ell}(c)(1 - p_{\ell}(c)) \leq A_S(\nu) \prod_{\ell \in S} p_{\ell}(c)$ , and  $\sum_{S \neq \emptyset} A_S(\nu) \leq R_1(\nu)$ .*

*Proof.*  $d_S(c) \prod_{\ell \in S} p_{\ell}(1 - p_{\ell}) = \mathbb{E}[\nu_c y^S] = Q_0 \mathbb{E}_n[\nu(n) g(n)]$  with  $g = \mathbf{1}[n \equiv c \pmod{Q_0}] \prod_{\ell \in S} y_{\ell}(n \bmod \ell)$ . By CRT,  $\hat{g}$  is supported on  $\Theta_S$  with  $|\hat{g}| \leq Q_0^{-1} \prod_{\ell \in S} p_{\ell}(c)$ ; apply Parseval. The sets  $\Theta_S$  are pairwise disjoint and avoid 0.  $\square$

**Theorem 12.3** (sieve limit with per-frequency rounding; [NC, Thm 2.3, Cors 2.4, 2.5]; [PROVED]; Case A [PROVED MOD] Elsholtz–Tao Prop. 1.4). *Take a finite prime-slice system with  $p_\ell(c) \leq 1/4$ , weights  $s_\ell = \log(1/(2p_\ell^+))$ ,  $p_\ell^+ = \max_c p_\ell(c)$ , and a majorant  $\nu$  of arbitrary level. Let  $\bar{\Phi}(\lambda, \alpha)$  be the right side of (4) minus  $\log(Q_0/|R|)$ , computed with these weights and  $s_* = \log 2$ . Then*

$$\mathbb{E}\nu \geq \frac{|R|}{Q_0} [(1 - \varepsilon)e^{-\bar{\Phi}(\lambda, \alpha)} - 2\varepsilon], \quad \varepsilon = \sum_{s(S) > \lambda} A_S(\nu)e^{-s(S)} \leq e^{-\lambda} R_1(\nu).$$

Consequently, for the families of Corollary 6.1, every bound  $\#(\mathcal{A} \cap [1, N]) \leq N\mathbb{E}\nu + R_w(\nu)$  with  $w \geq 1$  has saving at most  $C(\log N)^{3/4} + \log(P/\varphi(P))$ , with no bound on the size of the slice primes.

*Proof sketch.* In each fibre apply Proposition 12.1; by Lemma 12.2 both tails are at most  $\varepsilon$ . Average over  $c$  as in Theorem 4.5. If the bound equals  $Ne^{-s}$  then  $R_1 \leq N$ , and for  $N \geq N_0(C)$  the level  $\lambda = 2\log(4N)$  gives  $\varepsilon \leq e^{-\bar{\Phi}}/4$ . For the families of Corollary 6.1,  $s_\ell \geq \frac{1-C}{2} \log \ell$  for  $\ell \geq \ell_0(C)$ , so the profile is  $\ll \alpha^{-3}$  as there.  $\square$

Only the Fourier mass above level  $\lambda$  enters. This removes, for finite prime-slice families, the proviso “slice primes  $\leq N^{O(1)}$ ” (item (A3) of Definition 3.3), and it covers the coefficient sum, the sawtooth bound in the form  $\min(N, 1/(2\|\theta\|))$ , and complete exponential sums over the classes of one frequency. It does *not* cover [NC, §§2.3–2.5]: Erdős–Turán/Vaaler rounding with the factor  $2|\sin \pi N\theta|$ , per-class floor/ceiling rounding, the exact weight  $|S_N(\theta)|$ , smooth windows, or Kloosterman/dispersion cancellation over the moduli; these have weight  $< 1$  or cancel between frequencies. For smooth windows,  $Q_0 = 1$  and hit-pattern majorants, the cap follows from an equidistribution statement for sums  $\sum_{\ell \in S} h_\ell/\ell$  near 0, Conjecture ( $H_{\text{eq}}$ ) of [NC, (2.6)] [CONDITIONAL]; the other cases are [OPEN].

**12.2. Prime-only majorants.** A *prime majorant* is  $\geq 0$  at all primes and  $\geq 1$  at the primes of  $\mathcal{A}$ ; the bound is  $\sum_{p \leq N} \nu(p)$ .

**Theorem 12.4** ([NC, Lemma 3.1, Thms 3.2, 3.3]; [PROVED]). (1) *Assume  $|F_\ell(c) \setminus \{0\}| < \ell - 1$ . By Dirichlet’s theorem, a prime majorant is  $\geq 0$  on every reduced class modulo a common period  $L$  and  $\geq 1$  on every reduced class contained in  $\mathcal{A}$ . Theorem 4.5 and Lemma 3.4 hold for the uniform law on  $(\mathbb{Z}/L)^\times$ , with  $\varphi(Q_0)$ ,  $R \cap (\mathbb{Z}/Q_0)^\times$  and  $p_\ell^*(c) = |F_\ell(c) \setminus \{0\}|/(\ell - 1)$  in place of  $Q_0, R, p_\ell(c)$  (the budget lemma up to a factor  $\ll \log \lambda$ ). For the families of Corollary 6.1 with a selector, the cap  $C(\log N)^{3/4} + \log(P/\varphi(P))$  at level  $A \log N$  is unchanged.* (2) *If an integer majorant is  $\geq 1$  only on  $\mathcal{A} \cap \{(n, P(z)) = 1\}$  with  $z \leq e^\lambda$ , has level  $\leq \lambda$  in the system augmented by the class 0 at each  $\ell \leq z$  (the primes  $\ell < 5$  go into the selector), and the augmented probabilities are  $\leq 1/4$ , then the bound of Theorem 4.5 grows by at most  $C_4 \sum_{\ell \leq z} \ell^{-1-\alpha} + O(\log^2 \lambda)$ , i.e. by  $O((\log \log N)^2)$  at  $\alpha = \lambda^{-1/4}$ ,  $\lambda \asymp \log N$ .*

So prime equidistribution at level  $N^{O(1)}$  (Siegel–Walfisz, Bombieri–Vinogradov, Bombieri–Davenport–Halberstam, Elliott–Halberstam, GRH) cannot lift a prime majorant above the cap. That such majorants live at level  $N^{O(1)}$ , and that unconditional prime error terms are weaker than the integer count at relative accuracy  $e^{-(\log N)^{3/4}}$ , is an *Assessment* [NC, Remark 3.4], not a theorem.



**12.3. Moment methods.** For a finite witness family  $\mathcal{W}$  of classes put  $f(n) = \#\{W \in \mathcal{W} : n \in W\}$ . If  $P$  is a real polynomial with  $P \geq 0$  on  $\mathbb{Z}_{\geq 0}$  and  $P(0) \geq 1$ , then  $P \circ f$  is a CRT majorant of  $\{f = 0\}$  [NC, Prop. 4.1] [PROVED]. Hence integer moment methods of any order, whose moments are evaluated as  $N \cdot (\text{CRT density})$  plus a per-frequency error, are capped by Theorem 12.3; prime moments are capped at level  $\leq A \log N$  by Theorem 12.4, and beyond that level only under the Assessment above. For slice witness classes and  $\deg P \leq k$ ,  $\log(1/\mathbb{E}[P \circ f]) \leq 19k(2 + \log^+(C_4 \bar{\mu}/(19k))) + O(\log(\bar{\mu} + k)) + \log(Q_0/|R|)$  [NC, Prop. 4.2] [PROVED]; with witness moduli  $\leq N^{O(1)}$ ,  $\bar{\mu} \ll (\log N)^3$  and a degree- $k$  method saves  $O(k \log \log N)$  in the CRT mean, so the second moment gives only a power of  $\log N$ .

**12.4. Direct interval counts.** What is left of signed rounding is cancellation *between* frequencies in (9), i.e. a direct count of  $\sum_{n \leq N} \nu(n)$ . Theorem 12.3 turns into a dichotomy [NC, Thm 8.1] [PROVED]: if  $\sum_{n \leq N} \nu(n) \leq N e^{-s}$  then, for every  $\lambda$  and  $\alpha$ , either the Fourier mass above level  $\lambda$  satisfies  $\varepsilon > e^{-\tilde{\Phi}(\lambda, \alpha)}/4$ , or  $N \mathbb{E} \nu - \sum_{n \leq N} \nu(n) \geq N[(|R|/(4Q_0))e^{-\tilde{\Phi}} - e^{-s}]$ . For the families of Corollary 6.1 and  $s = (\log N)^\theta$ ,  $\theta > 3/4$ , this holds at every level  $\lambda \leq c(\log N)^{4\theta/3}$ , which is  $\gg \log N$ . Two facts about  $[1, N]$  are proved. Case-B classes of modulus  $> M_0 := 8[(N+1)/3]^2$  contain no  $n \leq N$ , so for hit-pattern majorants they give neither sieving power nor cost [NC, Lemma 8.2, Cor. 8.3]; and the multiplier-one witnesses of all moduli have mean  $\leq \frac{1}{4}(1 + \log N)^2$  on  $[1, N]$  [NC, Prop. 8.4]. The moduli in  $(N, M_0]$  still carry real hit mass  $\asymp (\log N)^3$  per integer, a constant factor below the CRT value [EVIDENCE]; a conjecture of [NC] to the contrary was withdrawn after review. For hit-count polynomials of degree  $k$  evaluated exactly on  $[1, N]$ , beating  $3/4$  needs  $k \geq (\log N)^{3/4+o(1)}$ , [CONDITIONAL] on a local-limit hypothesis  $H_{\text{node}}$  for the witness count, which is untested [NC, Prop. 8.6, Cor. 8.7]. Numerically, for polynomials of degree  $\leq 10$  in the hit count of a prime-modulus subfamily, the exact interval optimum saves less than the CRT optimum [EVIDENCE] [NC, §8.3].

### 13. WHAT THE THEOREMS EXCLUDE

The following list is checked against the proofs of Theorems 4.5, 4.9, 7.6, 9.1, 9.3 and Lemma 3.4 [ET, §6.1], [KA, §4.3], [NC, §6]. Anything below lies outside the cap of Theorem 9.4; where a later section covers part of an item, this is said.

- (1) *Majorants not  $\geq 0$  on all of  $\mathbb{Z}$ .* See Remark 3.6. Majorants that are  $\geq 0$  and  $\geq 1$  only at primes are covered for finite prime-slice families (Theorem 12.4).
- (2) *Signed rounding.* Bounds exploiting cancellation in  $\sum_{n \leq N} \nu(n) - N \mathbb{E} \nu$ , or rounding bounds smaller than  $\sum_i |a_i|$ . For finite prime-slice families, every per-frequency rounding bound with weight  $w \geq 1$  is covered (Theorem 12.3). Still outside: weights  $< 1$  (Erdős–Turán/Vaaler, floor/ceiling, the exact  $|S_N|$ , smooth windows), cancellation between frequencies, i.e. direct interval counts, and per-frequency rounding in the sequential and balanced settings of Sections 7–9.
- (3) *Non-CRT inputs.* The theorems see only the CRT law of residues. Covered for finite prime-slice families: prime-only majorants at level  $N^{O(1)}$  and CRT-evaluated moment methods (Section 12). Outside: Type I/II sums, the arithmetic of  $(p+a)/4$ , Halász-type arguments, moments counted as tuples of solutions rather than through CRT densities, and prime moments of



unbounded order beyond level  $A \log N$  (covered only under the Assessment of [NC, Remark 3.4]).

- (4) *Moduli without the  $B$ -hypothesis, and mixed families.* Theorem 9.3 covers every  $\mathcal{R}(M)$ -family with  $M \leq P(M)^{1+B}$ ,  $B$  fixed, twin moduli included; there is no summation over  $B$ , and the constants  $W_0(B), C(B)$  are not tracked. Excluded are:  $\mathcal{R}(M)$ -families with  $\log M / \log P(M)$  unbounded;  $(a, D)$ - and Case-A moduli without a dominant prime; and families mixing such classes with  $\mathcal{R}(M)$ -classes, since the base  $R_W$  rests on Lemma 2.6, proved for  $\mathcal{R}(M)$  only. For Selberg-type majorants at level  $O(\log X)$  the  $B$ -hypothesis is removed for any fixed number of prime factors above  $(\log X)^8$  (Theorem 11.12), and moduli with at least  $330 \log L$  of them cost  $o(1)$ . The  $\mathcal{R}(M)$ -moduli with large twin parts are not small (Lemma 13.1); for fixed  $B$  they are now covered.
- (5) *Non-selector admissible sets.* The term  $\log(Q_0/|R|)$  is uncontrolled (Remark 4.7). Corollary 6.2 handles the  $w_0$ -smooth part only because  $w_0 = O_C(1)$ , and Theorems 7.6 and 9.3 because  $W = W_0(B)$ .
- (6) *Level and size.* Families with slice primes beyond  $N^{O(1)}$  are formally outside Theorem 9.4, since Lemma 3.4 needs  $\Lambda_0$ . For finite prime-slice families Theorem 12.3 needs no such bound.
- (7) *Large multipliers.* In the prime-slice setting the small part  $q_0$  of a modulus is free, but Corollary 6.1 needs  $q_0 \leq \ell^C$ . Conditions with  $q_0 \gg \ell$ , used without paying their lcm in level, are not covered; no lower bound on that cost is proved [ET, §5.6(b)]. (Theorem 9.3 charges every prime above  $W$ ; it allows cofactors up to  $P(M)^B$  but does not address this item.)
- (8) *Large sieve.* Only Montgomery's large sieve over a slice system, fibre by fibre, is covered.

**Lemma 13.1** (balanced moduli carry cubic supply; [ET, Lemma 3.8]; [PROVED]). *There are  $c > 0$  and  $x_0$  such that for  $x \geq x_0$ ,*

$$\sum_{M \leq x, \substack{M \equiv 3 \pmod{4}, \\ P(M) \leq M^{1/2}}} \frac{|\mathcal{R}(M)|}{M} \geq c(\log x)^3.$$

*More precisely, for each fixed  $\eta \in (0, 1/480]$ , the moduli  $M = k\ell_1\ell_2$  with primes  $\ell_1 \in (Y, Y^{1+\eta/2}]$ ,  $\ell_2 \in (Y^{1+\eta/2}, Y^{1+\eta}]$ , odd  $k \in (Y^{2\eta}, Y^{3\eta}]$  and  $Y = x^{1/(2+4.5\eta)}$  already contribute  $\geq c(\eta)(\log x)^3$ ; these are same-scale pairs,  $\log \ell_2 / \log \ell_1 \leq 1 + \eta$ .*

*Proof sketch.* Each such  $M$  is  $\leq x$  and balanced, and determines  $(k, \ell_1, \ell_2)$ . For coprime  $u, v \leq z = Y^{1/8}$  with  $(uv, k) = 1$  and  $4uv \mid M+1$ , the class  $-uv^{-1} \pmod{M}$  lies in  $\mathcal{R}(M)$ , and distinct  $(u, v)$  give distinct classes modulo  $\ell_2$ . The  $\ell_2$ -sum over the progression  $-(k\ell_1)^{-1} \pmod{4uv}$  is evaluated by partial summation, with the error terms controlled on average by Cauchy–Schwarz against Brun–Titchmarsh and the Bombieri–Vinogradov theorem [D, Ch. 28]. The  $(u, v)$ -sum is bounded below by the lattice lemma of [LL, Lemma 3.1], which needs  $\eta \leq 1/480$ . The  $k$ - and  $\ell_1$ -sums give  $\gg \eta \log Y$  and  $\gg \eta$ . Full details are in [ET, §3.8], checked in [ETr, item 4].  $\square$

The share of the weighted Case-B mass carried by balanced moduli is 0.098, 0.121, 0.141, 0.157 and 0.171 at  $x = 10^3, \dots, 10^7$ , still growing [EVIDENCE]. Lemma 13.1 does not by itself show that every window partition loses: singleton windows have no internal pairs. So the twin moduli had to be covered, not discarded. For fixed  $B$

they cannot beat the cap (Theorem 9.3) [PROVED]; without the  $B$ -hypothesis this is open for general majorants.

#### 14. OPEN PROBLEMS

A proof of  $\theta > 3/4$  by a congruence sieve must therefore contain at least one of the following [ET, §6], [KA, §4.3], [NC, §6]:

- (i) moduli outside Theorem 9.4:  $\mathcal{R}(M)$ -moduli with  $\log M / \log P(M)$  unbounded, or  $(a, D)$ - and Case-A moduli without a dominant prime (alone or mixed with  $\mathcal{R}(M)$ -classes), used jointly in a way that beats their mass;
- (ii) multipliers larger than the slice primes, used without paying their lcm in level;
- (iii) a non-selector admissible set with a large saving of its own;
- (iv) a non-CRT input outside Section 12: rounding with weights  $< 1$  or cancellation between frequencies (a direct count of  $\sum_{n \leq N} \nu(n)$  in one of the two branches of the dichotomy of Section 12), counts of high-order correlations of solutions without CRT, or such inputs in the sequential and balanced settings.

Unresolved  $\eta$ -twin moduli, which formed item (i) in the first version of this list, are no longer on it for fixed  $B$ : Theorem 9.3 [PROVED]. For Selberg-type majorants at level  $O(\log X)$ , (i) is closed for  $\mathcal{R}(M)$ -moduli with a fixed number of prime factors above  $(\log X)^8$ , with or without the  $B$ -hypothesis (Theorems 11.11 and 11.12).

*Removing  $B$  for general majorants* [OPEN]. A transfer of the smooth-dominated sums of Theorem 11.12 to the block moments of Lemma 9.2 is sketched in [TW4, §11]; it would give  $\ll \lambda^{3/4} \log \lambda$  for every  $\mathcal{R}(M)$ -family. It is a *sketch* only, not checked, and not claimed here. The base (Lemma 2.6) and the moment lemmas are also what is missing for  $(a, D)$ - and Case-A classes; Theorem 8.5 itself is arithmetic-free [KA, §4.3].

The following statement would close (i) and (iii) in general.

**Hypothesis 14.1** ( $H_{\text{MS}}$ ; [ET, §5.6]; [OPEN]). For every CRT system of forced classes with moduli  $\leq e^\lambda$ , every majorant of level  $\lambda$  satisfies  $\log(1/\mathbb{E}\nu) \leq C \inf_\alpha [\alpha\lambda + \sum_{\text{cond}} \mathbb{P}(\text{cond})(\text{mod cond})^{-\alpha}] + O(\log^2 \lambda)$ .

Under  $H_{\text{MS}}$  every CRT-majorant architecture built from Case-B and Case-A forced classes, with a bounded-saving admissible set and rounding treated in absolute value, is capped at  $(\log N)^{3/4}$  [CONDITIONAL].  $H_{\text{MS}}$  is open even for one scale of primes with random pair conditions.

For the twin residual, the reduction of [TW, §6] isolated an arithmetic-free statement. Let  $V$  be a block of primes with costs  $s_\ell \in (s, (1+\eta)s]$  and  $d = \lfloor \lambda/s \rfloor$ .

**Conjecture 14.2** ( $k$ -ary comparison inequality; [TW, Conj. 6.4]; [OPEN]). Let  $\nu = \bigotimes \nu_\ell$  be a product law on  $V$ , and let  $\mathcal{F}$  be a family of hard  $k$ -ary constraints ( $2 \leq k \leq r$ ) on  $V$  with all incident weights  $\leq \theta \leq \theta_0(r)$  and total  $\nu$ -mass  $\mu_{\geq 2}$ . Let  $\sigma$  be the sequential law: in increasing order of  $\ell$ ,  $\nu_\ell$  conditioned off the residues activated by earlier coordinates, or  $\nu_\ell$  itself if that set has  $\nu_\ell$ -mass  $> \delta_\ell$ . Then for every  $\lambda$ -level  $f \geq 0$ ,

$$\mathbb{E}_\sigma f \leq \exp(C_r[d \log(2 + \mu_{\geq 2}) + 1]) \mathbb{E}_\nu f.$$

Here the *incident weight* of a residue  $a$  at  $\ell$  is the  $\nu$ -probability that some activated constraint through  $(\ell, a)$  is completed by the other coordinates.

The statement of Conjecture 14.2 is the *unweighted* form. Its weighted form, for the same law  $\sigma$ , is Theorem 8.5 [PROVED], and without the incident-weight hypothesis the unweighted form is false (Remark 8.7). With it, the unweighted form remains [OPEN], but it is no longer needed for the 3/4 question [KA, §5].

**Proposition 14.3** ([TW, Prop. 6.5]; [CONDITIONAL] on Conjecture 14.2 and Hypothesis K2). *Assume Conjecture 14.2 for  $r = \lfloor (1+B)(1+\eta) \rfloor$  and the  $k$ -ary second-moment hypothesis K2 of [TW, §6.3]. Then for every fixed  $B$ , every family of  $\mathcal{R}(M)$ -classes with  $M \leq P(M)^{1+B}$  (twin, prime-power top and all other shapes included), plus  $W_0$ -smooth classes, has saving  $\ll_B \eta^{-1} \lambda^{3/4} \log \lambda$ .*

Its conclusion, without the factor  $\log \lambda$  and without either hypothesis, is Theorem 9.3. Proposition 14.3 is kept only to record the route.

For  $\Lambda^2$ -majorants the core is a sparse noise-stability statement.

**Conjecture 14.4** (sparse noise stability; [TW, Conj. 6.8]; [OPEN]). Let  $X_i$  be independent and  $\mathcal{E}$  a finite family of events, each depending on coordinates  $S(E)$ , with incident weights  $\sum_{E \ni i} \mathbb{P}(E) e^{a|S(E)|} \leq \theta \leq \theta_0(a)$  and a codegree condition for  $|S(E)| \geq 3$ . Let  $\mathcal{A}$  be “no event” and  $\mathcal{A}'$  its  $\rho$ -correlated copy. Then

$$\log \frac{\mathbb{P}(\mathcal{A} \cap \mathcal{A}')}{\mathbb{P}(\mathcal{A})^2} \leq (1 + O(\theta)) \sum_E \mathbb{P}(E) \left( \prod_{i \in S(E)} (\rho_i + (1 - \rho_i) P_i(E)) - \mathbb{P}(E) \right),$$

where  $P_i(E)$  is the marginal factor of  $E$  at  $i$ .

Theorem 11.4 proves a bound of this type for binary systems ( $|S(E)| \leq 2$ ) with the diagonal and one-site terms on the right; the any-arity bound of [TW3, Lemmas 6.1, 6.2], with codegree terms and a free codegree quarantine, is what Theorem 11.11 uses. The conjecture itself is [OPEN].

*Many large primes in the  $\Lambda^2$  route* [OPEN]. The  $\Lambda^2$  caps of Section 11 are uniform for  $r \leq \epsilon \log L / \log \log L$  prime factors above  $(\log X)^8$ , and the moduli with  $r \geq 330 \log L$  are free. The middle range, which carries most of the  $\alpha^{-3}$ -weighted mass, needs per-prime-efficient bounds for the number of vertex hubs and for an off-diagonal pair sum [TW4, §12]. Under the  $B$ -hypothesis this concerns only the  $\Lambda^2$  method, since Theorem 9.3 covers every  $r$ ; without it, it is part of item (i).

*Non-CRT counts* [OPEN]. Conjecture ( $H_{\text{eq}}$ ) for smooth windows and the local-limit hypothesis  $H_{\text{node}}$  for exact interval counts (Section 12) are open, as is any treatment of non-symmetric hit-pattern majorants on  $[1, N]$  [NC, §8.6].

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